

LEBRE v0.53 — Specification and Technical Report

Online forecasting with statistically tested structural changes, a precision expert and a trivial-reference starting point, under a declared computational budget

LEBRE: Lifecycle-governed Evidence-Based Resource Evolution

Architecture: structural core (memory + structural expert + dynamic averaging, with changes tested by e-processes) + M1 (precision expert) + M2 (trivial reference)

STATUS

research version promoted by a pre-registered rule (2026-10-09); scope: one-step-ahead forecasting with inputs

FINAL EVALUATION

104 reserved series; $\text{LEBRE} \div \text{core alone} = 0.955$ [95% CI 0.951; 0.960]; no harm per family

COST

median of 919 operations per step (M1 + M2: +476, ceiling 1,100); estimated memory 83.1 KB

SCOPE OF THE EVIDENCE

M1 + M2 gain in basins and FX; identical to the core in solar, wind and load; core evaluated in the budget class on hydrology and buildings

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1. Executive summary

LEBRE is an **online** forecaster for time series with inputs: at every step it receives the inputs and, when available, the target value; it predicts the next value, learns from the error and moves on, without re-processing the past and without prior training. The per-step cost is fixed, given the number of inputs, and is counted operation by operation: over the 104 series of the final evaluation, a median of 919 and a maximum of 1,637 floating-point operations per step.

It has three layers (§4). The **structural core** is a simple live model (a memory of the target itself and a linear regression on the inputs) whose structure only changes by experiment: a challenger in shadow proposes to add, remove, swap or refine the response to a whole input, and the change is only accepted when an *e-process* — a sequential test valid at any time — accrues enough evidence of predictive improvement, within an online multiple-testing procedure. **M1** combines the core forecast with a precision expert (a recursive regression on the target's past and on lag-band means of the inputs). **M2** makes the forecast start from a declared trivial reference (zero, persistence or seasonal naive) and trust the adaptive part only when it proves better. None of these parts is new in isolation; the document describes the combination, the conditions under which it works and what was measured.

Premise: **LEBRE does not aim to be the most accurate forecaster.** The goal is to deliver useful forecasts, without severe failures, at a low and predictable per-step cost, without per-series tuning and with every structural change logged and justified. Expensive models are treated as **reference ceilings**, not as opponents.

What was measured

- **Pre-registered final evaluation** (104 series reserved before any M1 and M2 code; basins, buildings, solar and wind plants, electric load and FX): **LEBRE's** mean squared error divided by that of the core alone was **0.955** [95% CI 0.951; 0.960], with no harm in any family; the gain comes from Basins (CAMELS-BR) 0.879, FX: levels 0.952, FX: returns 0.966, Buildings (BDG2) 0.993; in solar, wind and load, **LEBRE** is identical to the core (§19).
- **Reference ceilings** in the same evaluation: against Chronos-2 with covariates (a foundation model of $\sim 10^9$ operations per forecast), a ratio of 0.88 to 1.43 per family — below 1 in basins and FX, above in buildings, solar, wind and load (§19).
- **The core in its budget class** (its own pre-registered evaluation, 60 basin and building series): error relative to online NLinear of 0.812 [0.752; 0.866], the lowest among the eight comparators of similar cost evaluated, with no series with a severe failure; a tie with a SARIMAX with inputs and $\sim 25\%$ higher error than Chronos-2 (§15).
- **Cost:** M1 and M2 add to the core a median of 476 operations per step (maximum 918; pre-registered ceiling 1,100); estimated memory 83.1 KB. The core, ported to C, ran on a **simulated** Cortex-M4F at 3.2–6.0 thousand instructions per step on average; M1 and M2 were not ported (§12, §16).

Main limits: one-step forecasting; M1 and M2 cost on every series and only gain in part of the domains; the core's testing mechanism **did not improve accuracy** compared to switching all inputs on without tests (ratio 0.988 [0.970; 1.006] on 100 series, exploratory) — its role is to produce a logged structure, whose practical usefulness was not demonstrated; inputs correlated with the true one may be accepted as substitutes; there is no formal error guarantee in the original scale for M1 and M2; the microcontroller was simulated and only for the core. No component is original (§21, §22).

2. The problem, the premise and the scope

The problem

Many systems continuously measure a target (river flow, building consumption, plant generation, electric load, an exchange rate) together with inputs that influence it, often with delay (rain, temperature, upstream flows, other currencies). Forecasting the next value *on the device itself* requires three things at once: very low per-step cost, operation without per-series tuning or prior training, and a behaviour that does not fail severely when the stream has gaps, changes or is nearly unpredictable. The most accurate modern models (foundation models) cost 10^7 to 10^{10} operations per forecast; classical models fitted series by series need maximum-likelihood optimisation; and cheap online linear models do not know, by themselves, which inputs and delays matter, nor when it is better to risk nothing beyond a trivial forecast.

The premise

LEBRE occupies a specific point of the accuracy–cost–transparency trade-off. It is evaluated **within its budget class** (models with up to $\sim 10^4$ operations per forecast), with the expensive models as ceilings. “Delivering well” means: competitive error within the class, **no catastrophic failure**, not losing to a declared trivial forecast, predictable cost and an explanation of every change.

Scope

- **one-step-ahead** forecasting, online, with the current inputs available (as in control and very-short-term forecasting);
- series with **few inputs** and an approximately linear relation, with delays of up to 31 steps in the core and up to 15 in the precision expert;
- the environment only provides the sampling period (e.g. 24 for hourly data with a daily cycle; for buildings, also 168 for the weekly cycle) and the trivial reference suited to the target (zero for changes such as financial returns, seasonal naive when there is a cycle, persistence otherwise);
- inputs standardised causally (exponential mean and variance), as an input contract.

What LEBRE is not

It is not a foundation model nor a long-horizon forecaster; it does not model strongly non-linear relations; it does not claim physical causality (an accepted input is an input with certified *predictive relevance* against the reference, and near-identical inputs are reported as a group). The name comes from the project’s original family of architectures (*Lifecycle-governed Evidence-Based Resource Evolution*): structure that enters and leaves the model by evidence, within a budget.

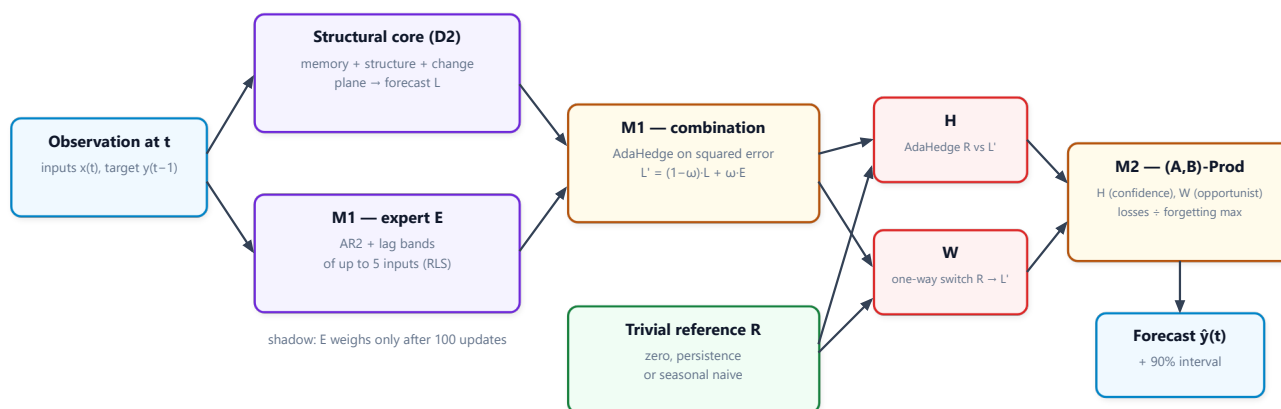
3. Design principles

Principle	Meaning
Every structural change is an experiment	Structure only changes when an anytime-valid sequential test rejects “the change does not improve the forecast”; each acceptance has an error level fixed beforehand.
Unit = a whole input	The question is “does this input help?”, not “does this lag help?”: fewer hypotheses, more power, the response being refined later by hierarchical splits.
Start from the trivial	The final forecast starts from a declared trivial reference and only comes to trust the adaptive part when it proves better on the data themselves.
Layers that do not touch the core	M1 and M2 only change how the final forecast is formed; the structural decisions are the same with or without them (verified on every evaluated series).
Fixed, declared cost	Fixed per-step cost given the number of inputs, counted operation by operation, with limits declared before each evaluation; the result is reported even when the limit is not met (§12).
No per-series tuning	The same constants on any series; the environment only provides the period(s) and the trivial reference.
Silence and out-of-contract data are not evidence	Without an observed target, with a clipped input ($ x > 8\sigma$) or in quarantine, nothing in the structure learns and no evidence accrues.
Output within contract	The core forecast never leaves an envelope of already observed values, and target gaps do not close the memory loop.
Observability	Every change is logged with time, kind, evidence and group of equivalent inputs; the weights of M1 and M2 are readable at every step; the forecast comes with an adaptive interval.

4. Architecture overview

D1 — Complete LEBRE architecture: structural core + M1 + M2

The core computes the forecast L and is the only owner of structural changes; M1 and M2 only change how the final forecast is formed



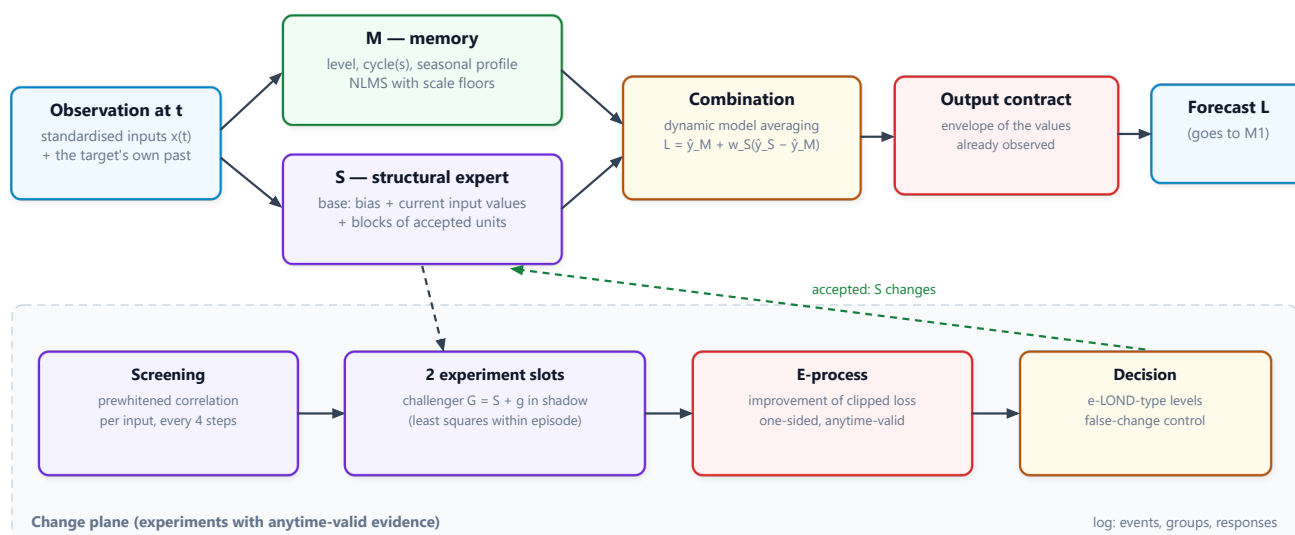
Complete view of one forecasting step. The core runs entirely underneath; M1 forms L' from L ; M2 forms the output from L' and the reference R .

LEBRE has three layers, executed in this order at every step:

1. **Structural core (D2)**: produces the forecast L and is the only owner of structural changes. It has two experts and a combination on the forecast path, and a change plane in shadow.
2. **M1 — precision expert (§9)**: a recursive regression E on the target's past and on lag-band means of the most correlated inputs, combined with L by AdaHedge on the squared error: $L' = (1 - \omega) \cdot L + \omega \cdot E$. Where E does not help, ω goes to zero and $L' = L$.
3. **M2 — starting from the trivial reference (§10)**: mixes R and L' through two paths — H , conservative (AdaHedge), and W , which switches from R to L' once (switch distribution) — and combines both with an (A,B)-Prod, which guarantees not falling far behind H .

D2 — Structural core: live forecast + shadow experiments

The forecast path (top) is cheap and always active; the change plane (bottom) proposes, tests and decides structural changes



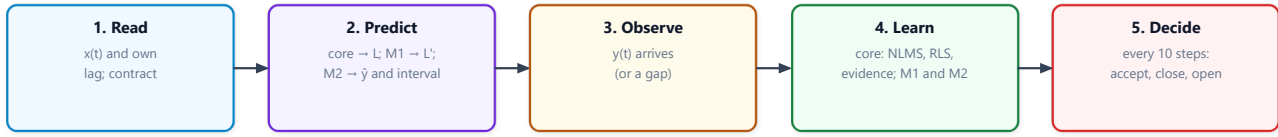
Structural core. The two experts and the combination form the path of the forecast L . The change plane runs in shadow and only alters S when a change is accepted.

Memory (M). Predicts the target from its own past, relative to the last value: reversion to a slow mean, the last difference, seasonal terms of the declared cycle(s) and a profile of increments per cycle phase. **Structural expert (S)**. Linear regression on the current value of each input (the “base”) plus the *response blocks* of the accepted units. **Combination**. Dynamic model averaging between M and S ,

with a weight that follows which of the two has been erring less. **Change plane.** A cheap screen points to the most promising candidate; up to two experiments run in shadow; an e-process accrues the evidence of each change; an online level procedure decides.

D3 — What happens in one time step

Strictly causal order: everything that decides the forecast at t uses only data up to $t-1$



Target gap: nothing learns; the memory receives the last observed value (hold), never its own forecast.

Order of operations in one step. The forecast at t is issued before $y(t)$ is read; M1 and M2 learn after the core, with the same $y(t)$.

5. Theoretical basis of the structural core

This section gathers the theoretical basis of the core. **All tools come from the literature** (cited in each item); what is specific to LEBRE is the way they are combined and the verified conditions of use (§8).

5.1 Online linear learning: NLMS and recursive least squares

The live model (base of S, block weights and memory) learns by **NLMS** (normalised least mean squares; Widrow & Stearns 1985; Haykin 2002):

$$w_{t+1} = w_t + \mu \frac{e_t}{\|\phi_t\|^2 + \delta} \phi_t, \quad e_t = y_t - w_t^\top \phi_t.$$

In steady state, the **misadjustment** (relative excess squared error caused by the weight jitter) is approximately $M \approx \mu / (2 - \mu)$. The core uses $\mu = 0.05$ in S and in the memory.

The **challengers** use recursive least squares (RLS) without forgetting within each episode, with $P_0 = 10 I$: with few parameters (5 to 7), RLS converges in tens of samples and gives an almost constant coefficient within the episode, which matters for the condition of use (§8). M1's precision expert also uses RLS, with forgetting (§9).

5.2 Memory: exponential smoothing in regression form

The memory is a regression of the increment on features of its own past, inspired by the *airline* model (Box & Jenkins 1970) and by the Holt-Winters seasonal state (Hyndman et al. 2008) — an analogy, not an equivalence:

$$\hat{y}_t^M = y_{t-1} + w_1(m_t - y_{t-1}) + w_2(y_{t-1} - y_{t-2}) + w_3(y_{t-s} - y_{t-1}) + w_4(y_{t-s+1} - y_{t-s}) + w_5 G_{t \bmod s} + [w_6(y_{t-s_2} - y_{t-1}) + w_7(y_{t-s_2+1} - y_{t-s_2})]$$

m: slow exponential mean (rate 0.01); G: exponential mean (rate 0.1) of the increments observed at the same cycle phase; s_2 : second declared cycle (week, for buildings). The NLMS normalisation has scale-relative floors (10^{-3} of the mean feature power and 10^{-4} of the slow scale of y), so that flat stretches do not produce huge steps.

5.3 Combination in the core: dynamic model averaging

The weight of S follows the discounted loss difference between the two experts (Raftery, Kárný & Ettler 2010; a logistic form equivalent to exponential expert weights):

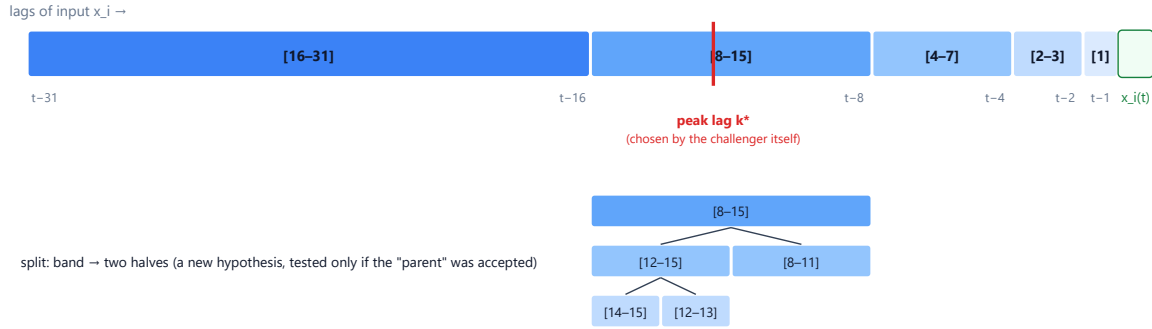
$$D_t = 0.99 D_{t-1} + \frac{(y_t - \hat{y}_t^S)^2 - (y_t - \hat{y}_t^M)^2}{\hat{\sigma}^2}, \quad w_S = \frac{1}{1 + e^{0.5 D_t}}, \quad L_t = \hat{y}_t^M + w_S(\hat{y}_t^S - \hat{y}_t^M).$$

5.4 Response representation: octave bands, peak lag and Haar splits

The response of an input is approximated by means of its past over dyadic bands [1], [2–3], [4–7], [8–15], [16–31] — a resolution that decreases with delay, as in multiresolution decompositions (Mallat 1989) and distributed-lag models. A band mean dilutes an isolated delay (it keeps $\sim 1/\text{width}$ of the effect); therefore the challenger also carries its **peak lag** k^* , chosen by itself as the largest prewhitened correlation in the first 200 samples of the warm-up — before any evidence, which keeps the challenger predictable. After acceptance, each band may be **split** into its two halves (one Haar coefficient); each split is a new hypothesis, tested only if the “parent” was accepted, in the spirit of hierarchical testing (Meinshausen 2008; Yekutieli 2008).

D4 — Unit of change: a whole input, refined by splits

An input challenger uses means of the past over octave bands; hierarchical (Haar-like) splits refine the response after acceptance



Representation of the response to an input: means over octave bands, the current value and a peak lag; after acceptance, each band may be split into halves, one hypothesis per split.

5.5 Screening: prewhitening and group statistic

To choose the next candidate without testing them all, each inactive input has a group statistic: exponential correlations (weight 0.02) between the residual of S and the band means, both **prewhitened** by a causal AR(1) of the input (Box & Jenkins 1970), squared, summed and normalised by the null variance. The unit is opened if the statistic exceeds the 99% quantile of χ^2_5 (15.09); a split, the 99% quantile of χ^2_1 (6.63), on the Haar detail. The screen **produces no evidence**: it only chooses what to test, and the error control comes entirely from the e-process.

5.6 Anytime-valid evidence: e-processes of predictive improvement

An **e-process** is a non-negative process whose expectation under the null does not exceed 1 at any stopping time; by Ville's inequality, the probability that it ever exceeds $1/\alpha$ is $\leq \alpha$. This allows looking at the evidence at every step, pausing and resuming without inflating the error (Ramdas, Grünwald, Vovk & Shafer 2023; Grünwald, de Heide & Koolen 2024). The core uses the sequential comparison of forecasters of Choe & Ramdas (2024), with the *weak* null: on average, the challenger does not improve the loss of the reference by more than a margin ϵ .

Clipped loss and differential, with the scale $B_t = 2 \max(\hat{\sigma}_t^S, 0.1 s_t^{\text{lenta}})$:

$$\ell_t(u) = \min\{(y_t - u)^2 / B_t^2, 1\}, \quad d_t = \ell_t(\hat{y}_t^S) - \ell_t(g_t) - \epsilon.$$

Sub-exponential mixture e-process, with predictable centring $\gamma_i = \min(\bar{d}_{i-1}, 0)$ and 8 values of λ :

$$E_n = \frac{1}{8} \sum_{j=0}^7 \exp \left\{ \lambda_j \sum_{i \leq n} d_i - \psi_{E,c}(\lambda_j) \sum_{i \leq n} (d_i - \gamma_i)^2 \right\}, \quad \psi_{E,c}(\lambda) = \frac{-\log(1 - c\lambda) - c\lambda}{c^2}, \quad \lambda_j = \frac{0.9 \cdot 2^{-j}}{c}.$$

The one-sided constant c . The statement of Choe & Ramdas assumes $|d_i - \gamma_i| \leq c$, which would give $c = 2(1+|\epsilon|)$. The underlying proof (Howard et al. 2021, §A.8) only uses the lemma of Fan, Grama & Liu (2015): $\exp\{\lambda \xi - \psi_E(\lambda) \xi^2\} \leq 1 + \lambda \xi$ for $\xi \geq -1$ and $\lambda \in [0, 1)$. For the one-sided test, under the weak null $E[d_i | \text{past}] = \mu_i \leq 0$, the **lower** bound $d_i - \gamma_i \geq -c$ suffices:

$$\mathbb{E}_{i-1} \left[e^{\lambda(d_i - \gamma_i) - \psi_{E,c}(\lambda)(d_i - \gamma_i)^2} \right] e^{\lambda(\gamma_i - \mu_i)} \leq (1 + \lambda(\mu_i - \gamma_i)) e^{\lambda(\gamma_i - \mu_i)} \leq 1.$$

Since $\lambda \Sigma d_i \leq \lambda \Sigma (d_i - \mu_i)$, the process is bounded by a non-negative supermartingale. Since $d_i \geq -(1+\epsilon)$ and $\gamma_i \leq 0$, $c = 1 + \max(\epsilon, 0)$. A mixture of e-processes is an e-process, and pauses between episodes form a predictable subsequence.

This observation is a direct reading of the published proof, not a new result. In a development simulation (400 trajectories $\times$ 5,000 steps, null at the boundary) the type-I error was ≤ 0.025 against $\alpha = 0.05$, and power rose from 0.66 (two-sided c) to 0.77 (§15). The betting alternative with predictable λ (Waudby-Smith & Ramdas 2024) had more power but accepted near-identical substitutes of the true input and was not adopted.

5.7 Multiplicity over the stream: e-LOND-type levels

Each change hypothesis receives, at creation, a fixed level that depends only on the number of hypotheses already created and of changes already accepted (the LOND rule of Javanmard & Montanari 2018, in the e-value version of Xu & Ramdas 2024). This controls the false change rate (FDR) over an endless stream, with asynchronous decisions:

$$\alpha_n = \min(\alpha \gamma_n (R_n + 1), 0.5), \quad \gamma_n = \begin{cases} 1/(2P) & n \leq P \\ \frac{1}{2} C / ((n - P) \log^2(n - P + 1)) & n > P \end{cases}, \quad \text{accept if } \log E \geq \log(1/\alpha_n),$$

with $\alpha = 0.05$, $P = 2 \times (\text{number of units})$ and C such that $\Sigma \gamma_n \leq 1$. A hypothesis is *persistent*: its evidence accumulates across episodes.

5.8 Adaptive interval

The 90% interval is a quantile of the absolute error updated by coverage feedback, in the spirit of adaptive conformal inference (Gibbs & Candès 2021): every 8 steps, $\hat{q} \leftarrow \max(0, \hat{q} + 0.1 \cdot \sigma \Sigma(\text{miss} - 0.1))$. The same rule is applied to the error of the final output (§11). There is no formal coverage guarantee under dependence; empirical coverage is reported.

6. Theoretical basis of M1 and M2: aggregation of forecasters

M1 and M2 are problems of **prediction with expert advice**: at every step two forecasts are available and one must decide how much weight to give each, using only the past. Three algorithms from the literature are used, without modifying their cores; LEBRE's choices are which losses they receive and how they are chained.

6.1 AdaHedge

Exponential weights on the cumulative losses with a learning rate tuned by the data themselves (de Rooij et al. 2014): it is the logarithm of the number of experts divided by the cumulative sum of the *mixability gaps* (how much the loss of the mixture exceeded the ideal “mix” loss). The regret against the best expert grows as the square root of the loss variation, and the algorithm is **invariant to translation and scale** of the losses: it does not need to know the error scale, which allows using the squared error without clipping or normalisation.

$$w_{k,t} = \frac{e^{-\eta_t S_{k,t-1}}}{\sum_j e^{-\eta_t S_{j,t-1}}}, \quad \eta_t = \frac{\ln K}{\Delta_{t-1}}, \quad \Delta_t = \Delta_{t-1} + \left(\sum_k w_{k,t} \ell_{k,t} + \frac{1}{\eta_t} \ln \sum_k w_{k,t} e^{-\eta_t \ell_{k,t}} \right),$$

with S_k the cumulative loss of expert k and $K = 2$. When $\Delta = 0$, the weights follow the leader.

6.2 Switch distribution with one switch

The *switch distribution* (van Erven, Grünwald & de Rooij 2012) is a Bayesian mixture over expert sequences that allows switching expert over time. LEBRE uses the simplest version: three hypotheses — “always R”, “always L” and “R until some time, L’ after” —, with equal priors and switch hazard $1/(n+1)$ at time n . The posterior is updated by the Gaussian likelihoods of the two forecasts, with a common variance equal to the smaller recent squared error of the two (with a floor). It concentrates weight quickly on the reference when the adaptive part does not help, and switches once when the adaptive part becomes clearly better. When all the mass is on one side, the direct update can give 0/0 by *underflow*; in that case the same computation is done in log scale.

6.3 (A,B)-Prod: not falling far behind a trusted expert

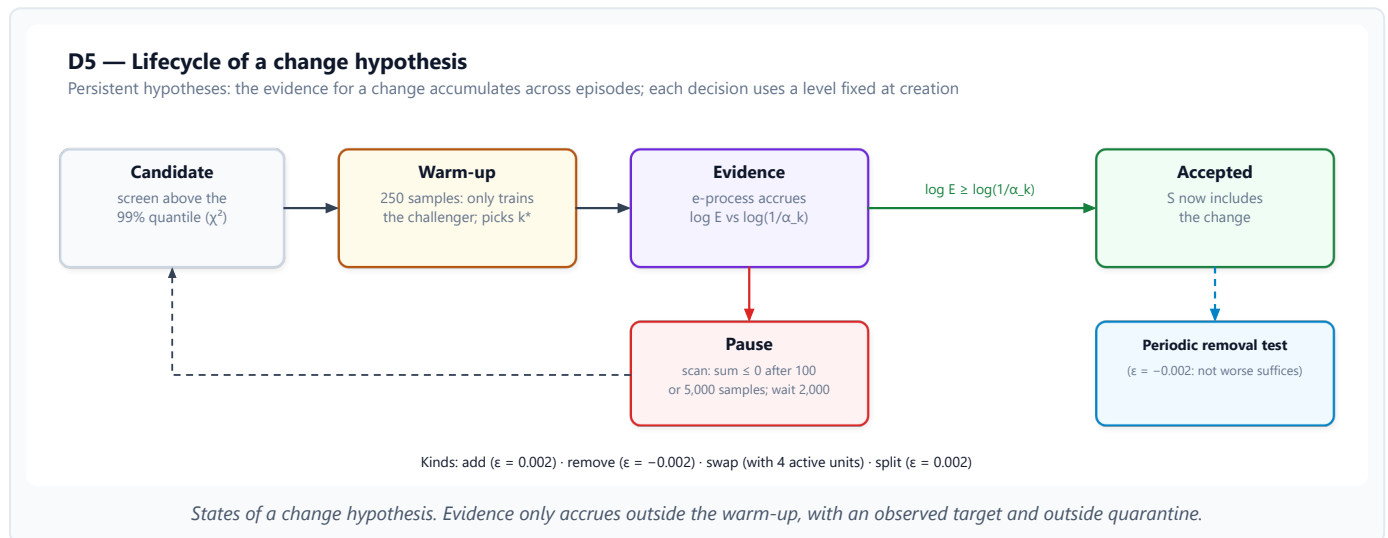
(A,B)-Prod (Sani, Neu & Lazaric 2014) combines an *opportunistic* algorithm A with a *trusted* algorithm B. B's weight is fixed at 1/2 and A's is updated multiplicatively by the loss difference; the fraction given to A is s_t . For losses in $[0, 1]$, it guarantees **constant** regret against B and of order $\sqrt{T \log \log T}$ against A, in the version without a known horizon. This kind of asymmetric guarantee — almost nothing to lose against the safe choice — comes from the Pareto frontier of regret (Even-Dar et al. 2008; Koolen 2013).

$$s_t = \frac{\eta_t w_t^A}{\eta_t w_t^A + w^B/2}, \quad w_{t+1}^A = w_t^A (1 + \eta_t \delta_t)^{\eta_{t+1}/\eta_t}, \quad \delta_t = \ell_t^B - \ell_t^A, \quad \eta_t = \min\left(\frac{1}{2}, \sqrt{1/(1 + \sum_{\tau < t} \delta_\tau^2)}\right).$$

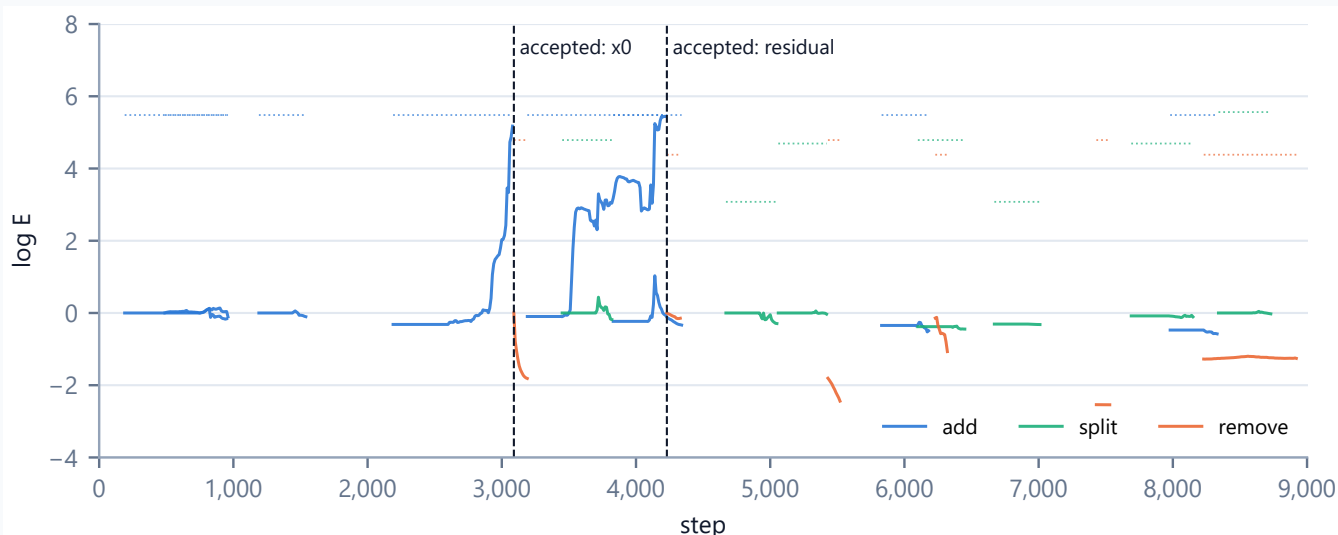
6.4 The limit of unbounded losses

The squared error has no upper bound. The known guarantees for unbounded and heavy-tailed losses (Cesa-Bianchi, Mansour & Stoltz 2007; Hazan & Kale 2010; Orabona & Pál 2018; Mhammedi, Koolen & van Erven 2019; V'yugin & Trunov 2019; Alquier 2021; Moulin, Esposito & van der Hoeven 2025) hold in units of the **largest single-step loss**. When a single step accounts for a large part of a series' whole error — as in an FX shock —, no online combiner can guarantee tracking the best expert with a small slack relative to the total error. This limit is a property of the data, not of the algorithm, and is handled explicitly in the evaluation (§10, §14).

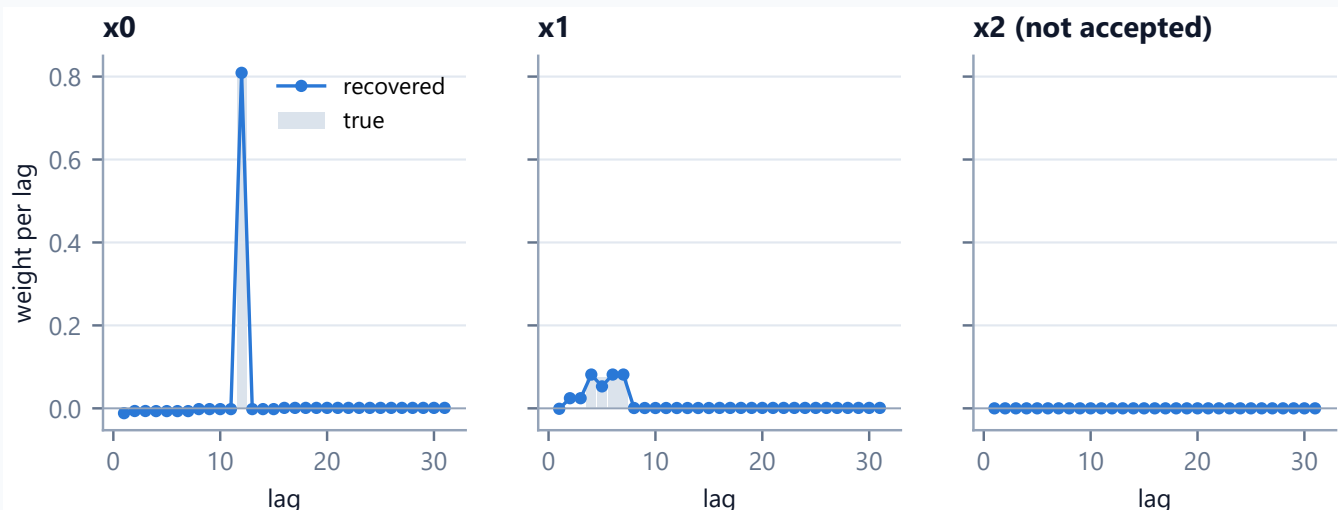
7. Lifecycle of a structural change



- **Units:** each input (including the target's own past, standardised causally) and the residual latent state (two filters of the standardised error, poles 0.8 and 0.95). Up to 4 active units.
- **Input challenger:** RLS on [means of the 5 bands, peak lag, current value $x_i(t)$] — the block is judged net of what $x_i(t)$ already carries. The RLS target is the residual the change would explain.
- **Episode:** 250-sample warm-up without evidence; ends on acceptance, by the scan rule (episode sum ≤ 0 after 100 samples) or at 5,000 samples; the same hypothesis waits 2,000 steps before returning. Only one challenger warms up at a time (cost-peak control).
- **Decisions** every 10 steps: accept, close, open. Priority: periodic removal test of active units; then the best screened candidate (unit or split). With 4 active units, the candidate enters by *swap* with the one of smallest contribution.
- **Judged on the forecast of S**, not on the combination nor on the final output: the question is whether the change improves the structural expert. M1 and M2 take no part in structural decisions.



Real example (development series: flow at a hydro plant; input x_0 = outflow of the upstream plant; x_1 = the target's own past). Log-evidence of the experiments (solid) against their thresholds $\log(1/\alpha_k)$ (dotted). Input x_0 is accepted at $t \approx 3,090$ and the residual latent state at $t \approx 4,230$; afterwards, removals and splits are tested and closed without evidence.



Declared synthetic process ($y = 0.8 \cdot x_0(t-12) + 0.3 \cdot \text{mean}(x_1, t-4 \dots t-7) + \text{noise}$; x_2 irrelevant). Response recovered by the core (dots) against the true one (bars): lag 12 of x_0 is recovered with weight 0.81; the distributed effect of x_1 appears in band [4–7]; x_2 is not accepted.

8. The condition of use: reference misadjustment

A test of predictive improvement compares **methods** (model + estimation procedure), not models — a distinction known in econometrics (Giacomini & White 2006). During development, this appeared concretely: with the base of S learning by NLMS with step 0.1, targets **without any structure** had a change accepted in ~20% of the runs. The cause is that the estimation error of the reference is predictable from the past, and a challenger can “correct” it.

Let $f_t^* = \mathbb{E}[y_t | \mathcal{F}_{t-1}]$, $\eta_t = f_t^* - S_t$ (estimation error of the reference, predictable) and the challenger $G_t = S_t + g_t$. For the normalised squared-loss difference $\Delta_t = [(y_t - S_t)^2 - (y_t - G_t)^2] / B_t^2$:

$$\textbf{Proposition 1} \quad \mathbb{E}[\Delta_t | \mathcal{F}_{t-1}] = \frac{2g_t\eta_t - g_t^2}{B_t^2} \leq \frac{\eta_t^2}{B_t^2}, \quad \text{with equality at } g_t = \eta_t.$$

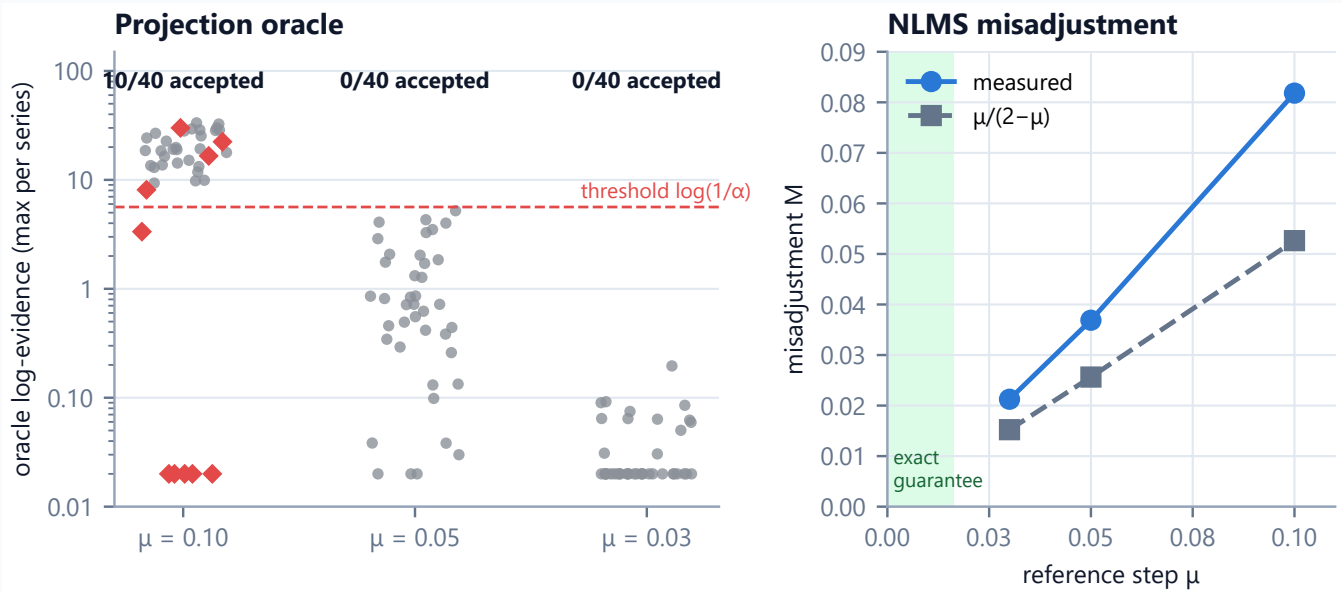
Proof: $y - S = \varepsilon + \eta$ and $y - G = \varepsilon + \eta - g$; expanding, the difference is $2g(\varepsilon + \eta) - g^2$, and $\mathbb{E}[\varepsilon | \text{past}] = 0$. ■

Corollary (exact guarantee) $\frac{1}{n} \sum_{i \leq n} \frac{\eta_i^2}{B_i^2} \leq \varepsilon \quad \forall n \Rightarrow$ under the structural null, the test null holds.

For NLMS with misadjustment M , this requires $M \lesssim 4\varepsilon$ (with $\varepsilon = 0.002$: $\mu \lesssim 0.016$).

Proposition 2 (fixed coefficient) mean improvement over the window $\leq D_u + \varepsilon$, $D_u = \frac{1}{|W|} \sum_{i \in W} (P_u \eta_i)^2 / B_i^2 - \varepsilon$,

where P_u is the projection of η on the unit's features. **Proposition 3 (practical criterion, approximation):** log-evidence grows as $n \cdot D^2 / (2\sigma_d^2)$; on a series of length T , false acceptances are unlikely if $D_u < \sigma_d \sqrt{2 \log(1/\alpha_k) / T}$.



Development check (120 runs with structureless targets, real inputs). Left: log-evidence of the best possible challenger (projection oracle) against the threshold; actual acceptances in red. With $\mu = 0.1$ the oracle crosses the threshold (median 18.2) and there were 10/40 acceptances; with $\mu = 0.05$ (max 5.2) and $\mu = 0.03$, none. In runs that accepted, the oracle was measured after the acceptance and may be low (points at the floor). Right: measured and theoretical misadjustment.

Honest reading: the core uses $\mu = 0.05$, which lies in the **practical regime** of Proposition 3 (the deviation from the null exists, ~ 0.0013 , but is smaller than what is detectable within the series length, ~ 0.0019), and **not** in the exact-guarantee regime ($\mu \lesssim 0.016$). Therefore false-change control is checked empirically with null batches (0/40 and 0/30 acceptances in development), not only asserted by theory. The observation relates to Giacomini & White (2006) and, through another mechanism, to Han & Qu (2026); we did not find it stated explicitly for this situation, but it is a simple consequence of known results.

9. M1: precision expert

Motivation. The core estimates the target dynamics (memory) and the input effect (structure) separately, with cheap learning. In some families this leaves it **below the accuracy × cost frontier**: a regression of equal or lower cost, estimating both jointly, errs 5% less (§17). M1 adds that joint estimation as a second expert, without touching the core.

The expert E

Variables: the target itself at lags 1 and 2 and, for each of the $m = \min(d, 5)$ inputs with the largest absolute correlation with the target (d = number of inputs), the means over the lag bands $[0]$, $[1]$, $[2-3]$, $[4-7]$ and $[8-15]$:

$$z_t = [1, \tilde{y}_{t-1}, \tilde{y}_{t-2}, \tilde{x}_{i,b}(t)]_{i \leq m, b \leq 5}, \quad k = 3 + 5m, \quad E_t = \text{clip}(w_t^\top z_t, L_t \pm 2 \max(\hat{\sigma}_L, \text{floor}))$$

- **Input selection:** exponential means (rate 0.01, past only) of x , x^2 , $x \cdot y$, y and y^2 , updated every 8 learned targets; the selection is made after 100 targets and redone every 5,000. The means and deviations at selection time standardise the chosen inputs and the target. If the set changes, the regression and the combination restart.
- **Missing inputs** are replaced by the last value seen (0 before the first).
- **Weights:** recursive least squares with forgetting $\lambda = 0.999$ and $P_0 = 100 \cdot I$, updated every 8 learned targets (outside quarantine and with inputs within the contract); the matrix is symmetrised at every update and reset if it loses positive definiteness. E only exists after k updates (determined regression).
- **Clip:** E stays within $L \pm 2 \cdot \max(\hat{\sigma}_L, \text{floor})$, with $\hat{\sigma}_L$ the recent root mean squared error of the core (exponential mean, rate 0.01) and the floor 0.1 times the slow scale of the target; this bounds the damage of an ill-conditioned regression.

The combination with the core

AdaHedge (§6.1) with two experts, L and E , on the **squared error without clipping or normalisation**: the weights follow the accumulated squared error, which is what the mean squared error measures, and the learning rate adapts to the scale by itself. The combination observes every observed target at which E exists. **Shadow entry:** until 100 updates, $L' = L$.

$$L'_t = (1 - \omega_t) L_t + \omega_t E_t, \quad \omega_t = \frac{e^{-\eta_t S_{t-1}^E}}{e^{-\eta_t S_{t-1}^L} + e^{-\eta_t S_{t-1}^E}}, \quad S_{t-1}^\bullet = \sum_{\tau < t} (y_\tau - \bullet_\tau)^2$$

Why AdaHedge and not an (A,B)-Prod: Prod needs losses in $[0, 1]$. Three ways of putting the squared error into that interval were measured and failed for different reasons (§17): clipping at 2σ leaves Prod blind to large errors; dividing by the largest error ever seen freezes it after a record; dividing by the largest recent error gives small weight to the large-error steps, exactly where the expert gains. Trading Prod's constant guarantee for AdaHedge's (a regret that, relative to the total error, vanishes with series length) is what worked.

10. M2: starting from the trivial reference

Motivation. In nearly unpredictable series (for example, FX returns and levels), an adaptive part may err more than predicting zero or repeating the last value: it learns noise. M2 makes the output start from a **trivial reference R** declared with the data — zero for changes such as financial returns, seasonal naive (the value one cycle ago) when there is a cycle, persistence (the last value) otherwise — and only come to trust L' when it proves better. While R is undefined (before a full cycle, in the seasonal case), the output is L' .

- **H** — AdaHedge between R and L' on the squared error (§6.1): confidence, no switches.
- **W** — one-way switch from R to L' (§6.2), on Gaussian losses $(y - \cdot)^2 / 2s^2$, with $s^2 = \max(\min(\sigma_R^2, \sigma_{L'}^2), \text{floor}^2)$ and σ^2 exponential means (rate 0.01) of each side's squared errors, past only.
- **Combination** — horizon-free (A,B)-Prod (§6.3) with $B = H$ (confidence) and $A = W$ (opportunistic); the losses are $(y - \cdot)^2 / N_t$, with $N_t = \max(\text{step losses}, 0.99 \cdot N_{\{t-1\}})$ — the largest recent error, with forgetting, to put the losses into $[0, 1]$ without freezing after a record.
- H , W and Prod only learn on steps with an observed target, outside target quarantine and with R defined.

$$\hat{y}_t = s_t W_t + (1 - s_t) H_t, \quad H_t = w_{R,t}^H R_t + w_{L',t}^H L'_t, \quad W_t = w_{R,t}^W R_t + w_{L',t}^W L'_t$$

Gate audit. In parallel, an e-process (§5.6) with its own budget $\alpha = 0.01$ and margin $\epsilon = 0.002$ tests whether the adaptive part improves on the current reference (and, after a switch, the opposite); each acceptance is logged as an event and the hypothesis is recreated. This test **does not decide the output**: it only documents when the evidence changed sides, with the same kind of guarantee as the structural changes.

Guarantees and the limit of shocks

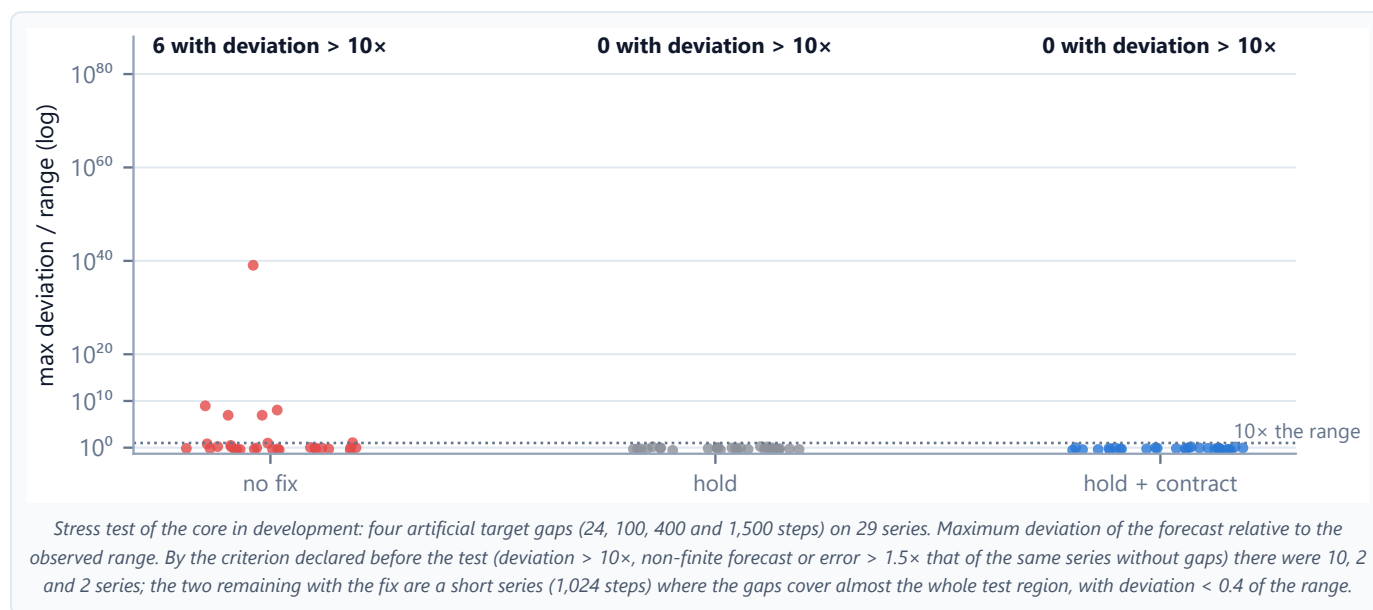
For the normalised losses, Prod guarantees constant regret against H and of order $\sqrt{T \log \log T}$ against W ; in the original scale there is no clean formal bound, because the normalisation changes over time. The limit of §6.4 shows up in the data: in financial series a single

day can account for much of the whole error (in the 521 development series measured, 32 have the largest step above 5% of the reference's total error; the largest, baht_1990, 25.9%). That is why the evaluation ruler treats those series as undecidable (§14).

11. Output safeguards and interval

Two safeguards act on the output path of the core and cost ~15 operations per step. Both are standard engineering practice; they are here because their absence produced the most severe failure observed (§15).

- **Hold during target gaps:** the memory receives the last observed value, never its own forecast. Without this, the memory runs open-loop and, with some learned weights, the recursion is unstable (characteristic root ~1.75 in a real case).
- **Output contract:** the core forecast is limited to the envelope of observed targets (running maximum and minimum relaxing toward the slow mean at rate 10^{-3}), widened by $0.5\times$ its width; a non-finite forecast becomes the last observed value.
- **M1 and M2 inherit the containment:** E stays at most $2\cdot\max(\sigma_L, \text{floor})$ from L; R is an already observed value (or zero); every output of M1 and M2 is a weighted mean of contained forecasts. A non-finite loss never enters the weights.



Output interval

The final output $\hat{y}$ comes with a 90% interval computed by the rule of §5.8 applied to the error of $\hat{y}$ (starting at 1.645 times the recent root mean squared error). The coverage of the core interval was measured in the core's own evaluations (§13); the coverage of the complete output interval **was not reported** in the final evaluation.

12. Computational cost and memory

Cost is counted operation by operation in the code (additions, products, divisions, roots, exponentials and logarithms count 1 FP), per component. The per-step cost is **almost fixed**: memory, base, screening, M1 and M2 always run; the core's experiments add cost only when a slot is busy, and peaks occur at decision steps. Nothing grows with time.

Core

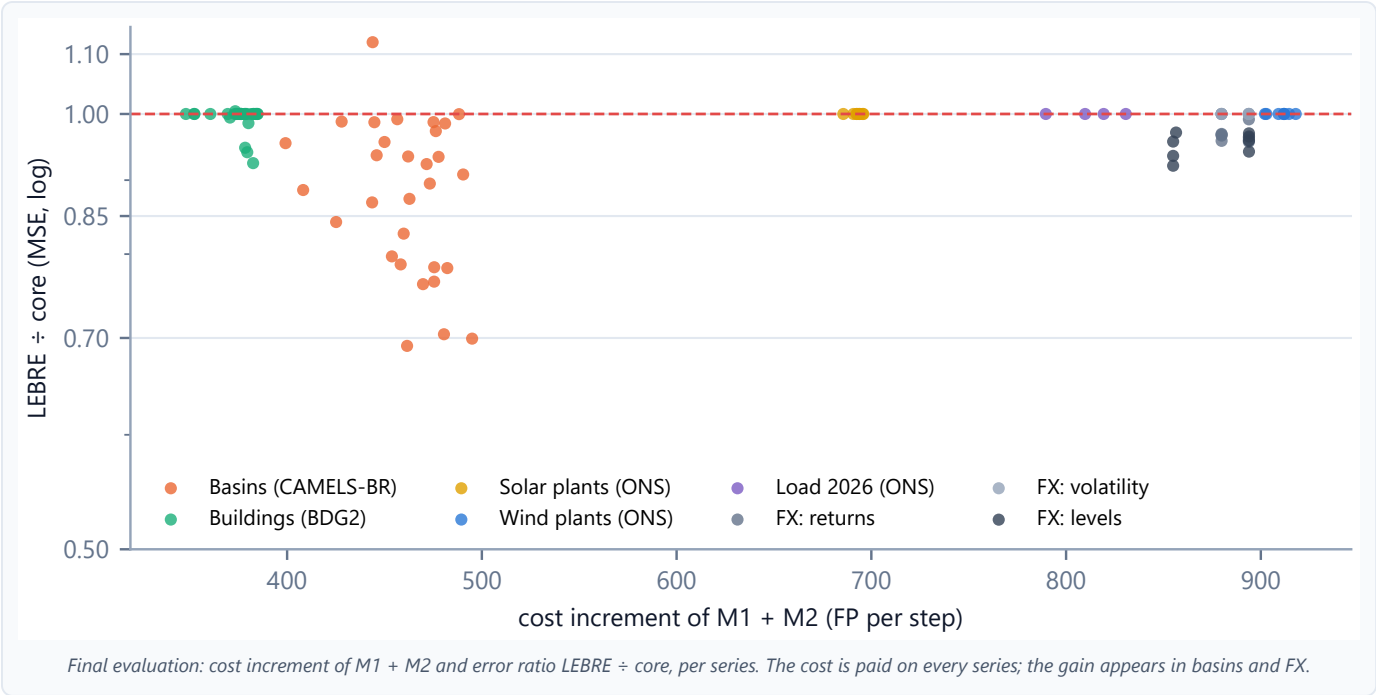
The core was designed with a contract of mean ≤ 400 and maximum $\leq 1,000$ FP per step on real series. In its own pre-registered evaluation (60 series), the mean was 415 and the maximum 1,020: the contract **was not met**, by 3.7% in the mean and 2.0% in the peak, and it was not revised after the result. In LEBRE's final evaluation, the core cost a median of 461 FP per step.

M1 + M2

The count of M1 and M2 was audited: the implementation performs exactly the declared operations (each pair of the RLS matrix symmetrised once; only the new input row standardised), with forecasts identical to the validated ones in 813 of 813 series; the audit raised the count by 3.2% (median). Per RLS update: $6k^2 + 5k$ (every 8 targets); expert prediction: $2k + 18m + 4$; AdaHedge: 48; M2: ~132 to 137.

Part	FP per step (median, one series per development family, before the audit)
core	453
M2	137
M1: RLS update	513
M1: rest	146
Total	1,257

Cost ceiling of M1 + M2. The preliminary design target (total cost ≤ 1.25 times that of the core) proved incompatible with M1 + M2: with the core at ~ 450 FP, the maximum increment would be ~ 110 FP, less than M2 alone. Before the final evaluation, the project owner replaced it with a ceiling of an **absolute increment of 1,100 FP per step** on every series (the new cost is nearly fixed; a ratio would penalise the series where the core is cheap), plus 2.5 times the core's cost on the families below the frontier. In development (813 series) the increment had a median of 538 and a maximum of 974 FP; in the final evaluation, a median of 476 and a maximum of 918 FP, with total cost between 667 and 1,637 FP.



Memory

The state has a fixed size, given the number of inputs and the periods. The core state was measured in the C port (capacities for up to 8 inputs and 160 hypotheses): 74,988 bytes; most of it is the hypothesis table, which can be reduced. M1 and M2 hold 1,263 numbers ($k = 28$ in the worst case: 1056 in M1, 207 in M2), 10,104 bytes in double precision. Total **estimated**: 85,092 bytes (83.1 KB), within the pre-registered limit of 128 KB; M1 and M2 were not ported nor measured.

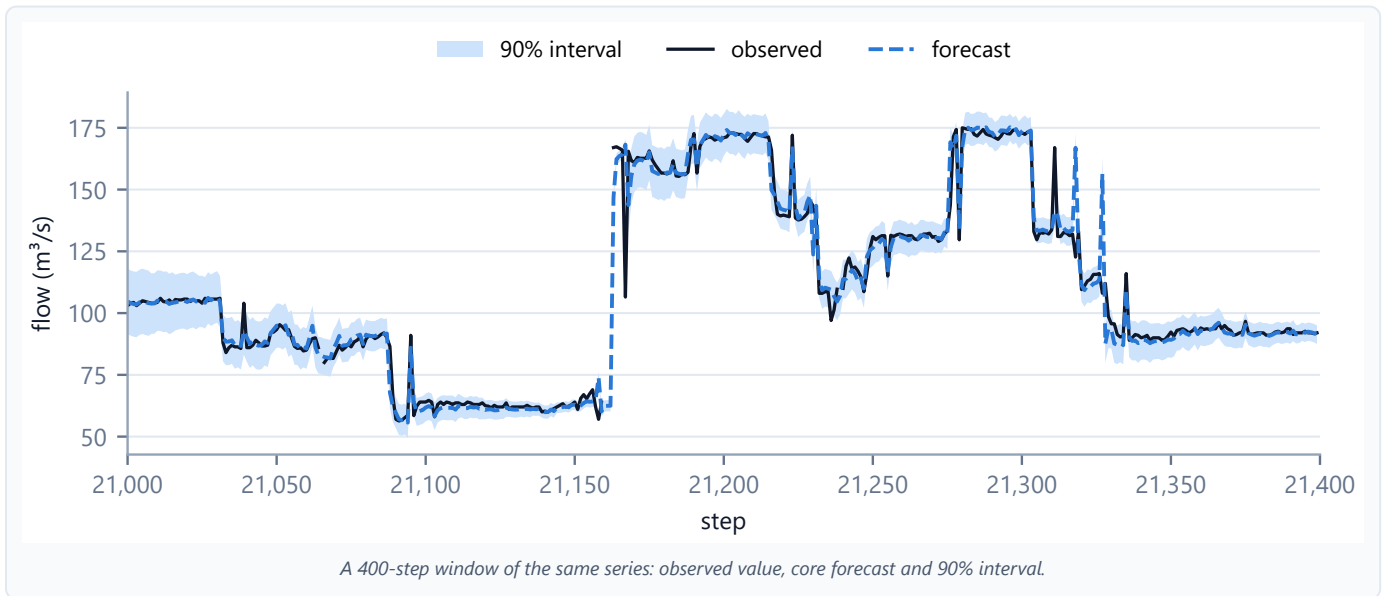
13. Observability and explainability

Each forecast can be decomposed into named contributions: in the core, memory, base and each active block; then, M1's weight ω and, in M2, the weights of R and L' in H and in W and the fraction s_t . Every structural change and every acceptance of the gate audit is logged. Example from a core development series:

Time	Event	Change	log E	Evidence samples
3,090	accepted	add input x0	7.18	957
4,230	accepted	add residual latent state	5.49	893

- Equivalent groups:** together with the accepted input x0 the model reports the group {x0, x1} (causal correlation ≥ 0.95 between x0, the outflow of the upstream plant, and x1, the target's own past). The claim is "this input or an equivalent one".
- 90% interval of the core:** empirical coverage of 0.906 on this series and 0.904 on average over the 60 series of the core's own evaluation.

- **Log of closed experiments:** each test that did not pass is logged with its evidence (for instance, removal tests of x_0 with log E between -1.8 and -3.2 : the input remains useful).
- **M1 and M2:** M1's final weight ω is read per series (§19: $\omega = 0$ on every solar, wind and load series); which inputs M1 chose is also available.



14. Evaluation methodology

Two questions were evaluated with their own protocols, both pre-registered: (a) is the core competitive within its budget class? (b) do M1 and M2 improve the core without harming any domain?

Common discipline

- **Online one-step forecasting**, without looking ahead; all models receive the same inputs, standardised causally.
- **Pre-registration**: before each access to reserved data, a document fixed questions, metrics, comparators, criteria and the code hashes; each reserve was run a single time; the data loader blocks access to the reserves outside a pre-registered run. Infrastructure incidents (without any model change) are described with the results.
- **Series split** by fixed-seed permutations, before any modelling; each validation set was chosen from metadata only, frozen with SHA-256 and used once.

(a) The core in the budget class

- **Metric**: mean squared error on the test region (last 70% of each series, observed target), divided by that of the calibrated online NLinear on the same series; geometric mean across series; 95% CI by bootstrap stratified by domain (2,000 resamples). **Catastrophic failure**: relative error > 10, deviation > 10× the observed range, or a non-finite forecast.
- **Data**: flows and generation from the Brazilian National Grid Operator (ONS; cascaded plants), CAMELS-BR basins (Chagas et al. 2020; precipitation, evapotranspiration, temperature) and Building Data Genome 2 meters (Miller et al. 2020; weather), plus two system-identification benchmarks in development.
- **Foundation models** (too expensive to run at every step): evaluated on 1,000 points per series, the same for all, with the same context and the current inputs as known future covariates. **Batch-fitted comparators** (SARIMAX with inputs, FITS, SparseTSF, TTM fine-tuning): fitted on the first 15% of the series (up to 5,000 points), then with fixed weights.

Set	Series	Use
Development	29 (ONS 11, CAMELS 10, BDG2 5, Silverbox, Cascaded Tanks ×2)	all design and tuning of the core
Reserve 1	125 (ONS 45, CAMELS 50, BDG2 30)	revealed the gap failure (§15)
Reserve 2	40 (CAMELS 20, BDG2 20)	checked the fix
Reserve 3	60 (CAMELS 30, BDG2 30)	ultralight comparators, TTM, C port, microcontroller

(b) M1 and M2 against the core

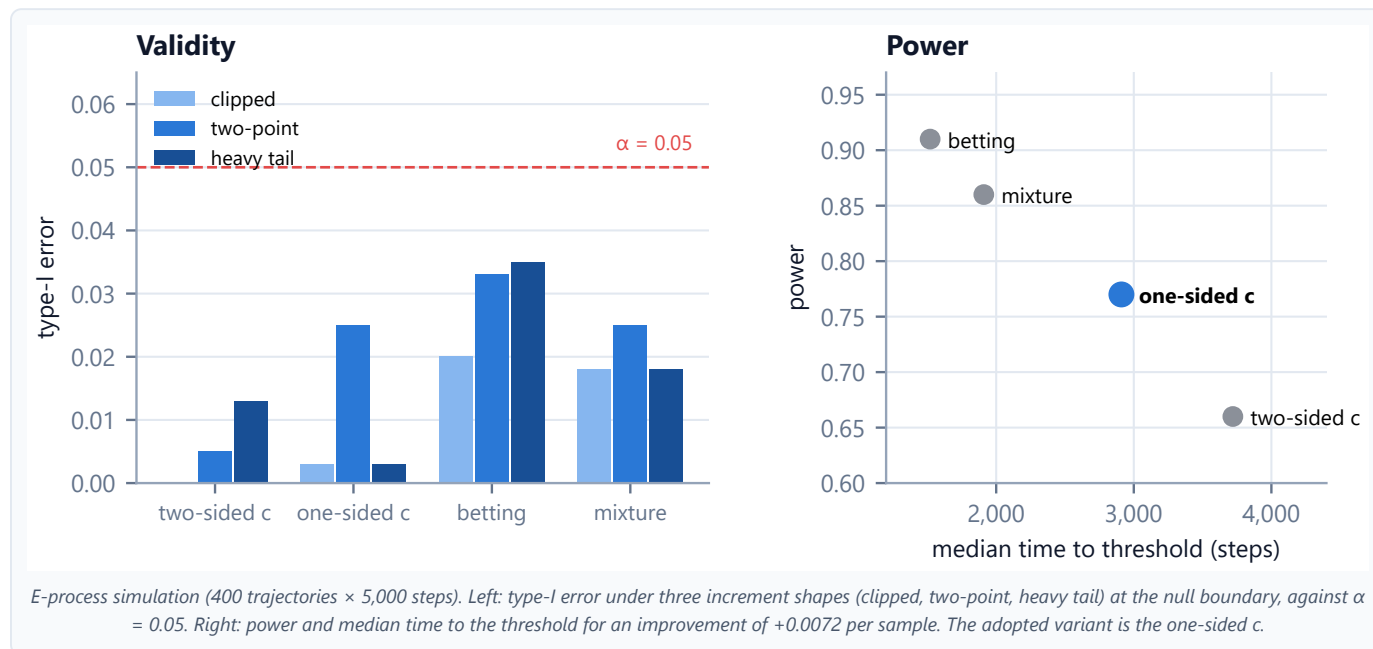
The final-evaluation data were reserved on 2026-10-05, before any M1 and M2 code, with hashes and locks. The metric is the mean-squared-error ratio $\text{LEBRE} \div \text{core}$, series by series. **Decidability ruler**: each series is classified by the 95% CI of $\text{core} \div R$ (paired block bootstrap, 2,000 replicates) as “core better”, “reference better” or “undecidable”. Criteria: (1) on “reference better”, $\text{LEBRE} \div R$ with upper limit ≤ 1.01 ; (2) on “core better”, $\text{LEBRE} \div \text{core}$ with upper limit ≤ 1.02 ; (4) on negative tests, no more false inputs than the core; (6) the core's structure preserved; (7) on short series (360 steps), ≤ 1.05 of the best of core and R . **Shock addenda** (decided during development, before validation 5 and the final evaluation): a series whose largest single step exceeds $\max(5\%, 5 \cdot 2 \ln(n)/n)$ of R 's error is undecidable on criteria 1 and 2 (n = evaluated steps), because on it no combiner has a useful guarantee (§6.4).

F6, on the families where the core lies below the accuracy × cost frontier: cost ≤ 2.5 times that of the core and error ≤ 1.05 times the lowest error among the diagnostic's 8 regressions with cost $\leq$ that of LEBRE. **Promotion** (pre-registration; project-owner decisions): (A) no harm per family, (B) $\text{LEBRE} \div \text{core}$ over all series with the upper limit of the 95% CI below 1 and (C) estimated memory ≤ 128 KB.

15. Evidence for the structural core

The core was developed and evaluated before M1 and M2 existed, with protocol (a) of §14. Since M1 and M2 do not alter any decision of the core (identical structure verified on every series of development, validations and the final evaluation), this evidence still holds for the core within LEBRE.

15.1 Development: validity and discovery

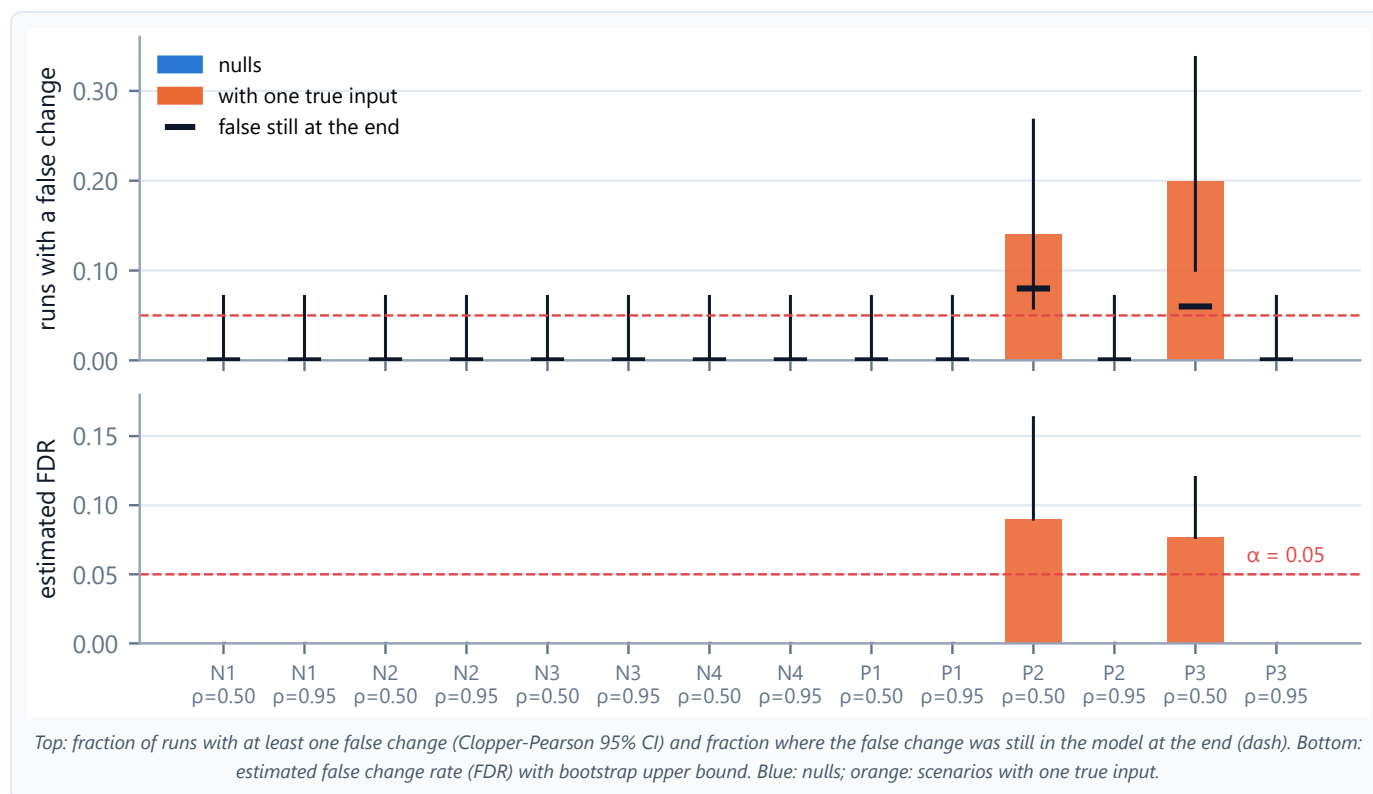


Situation	Result (development)
Synthetic with white inputs (delays up to 12)	structure recovered to the exact delay in 500–2,200 steps (e.g. delay 12, weight 0.84 vs 0.80)
Semi-synthetic (real inputs, known effect)	true input (or its group) in 2 of 3 structured tasks, 3/3 seeds; latency 4–17 thousand steps
Very smooth inputs (autocorrelation ~0.99), medium/long delay	not discovered ; an online oracle (knows the delay, tests from the start) needs 9–27 thousand steps or never gets there in 32 thousand
Structureless targets (null batches with real inputs)	0/40 and 0/30 acceptances; 0 false exogenous inputs in all declared sets
Near-identical inputs (correlation 0.98)	indistinguishable by predictive improvement; the model reports the group

The limit on smooth inputs is known in principle: with very persistent regressors, the new information of a delay beyond the current value is small (Stambaugh 1999; Campbell & Yogo 2006). In those cases the forecast remains good because the base already uses the current input values; what is lost is the *explanation* of the delay.

15.2 False-change control in simulation with dependence

The frozen core was run in 700 synthetic runs (plan and criterion declared beforehand; 4 mutually correlated inputs, persistence 0.5 or 0.95, 30 thousand steps): four nulls (white noise; autoregressive target; noise whose variance depends on an input; variance break) and three scenarios with one true input at delay 6 (plain; with a decoy of correlation 0.7 with the true one; with heavy-tailed autoregressive noise).

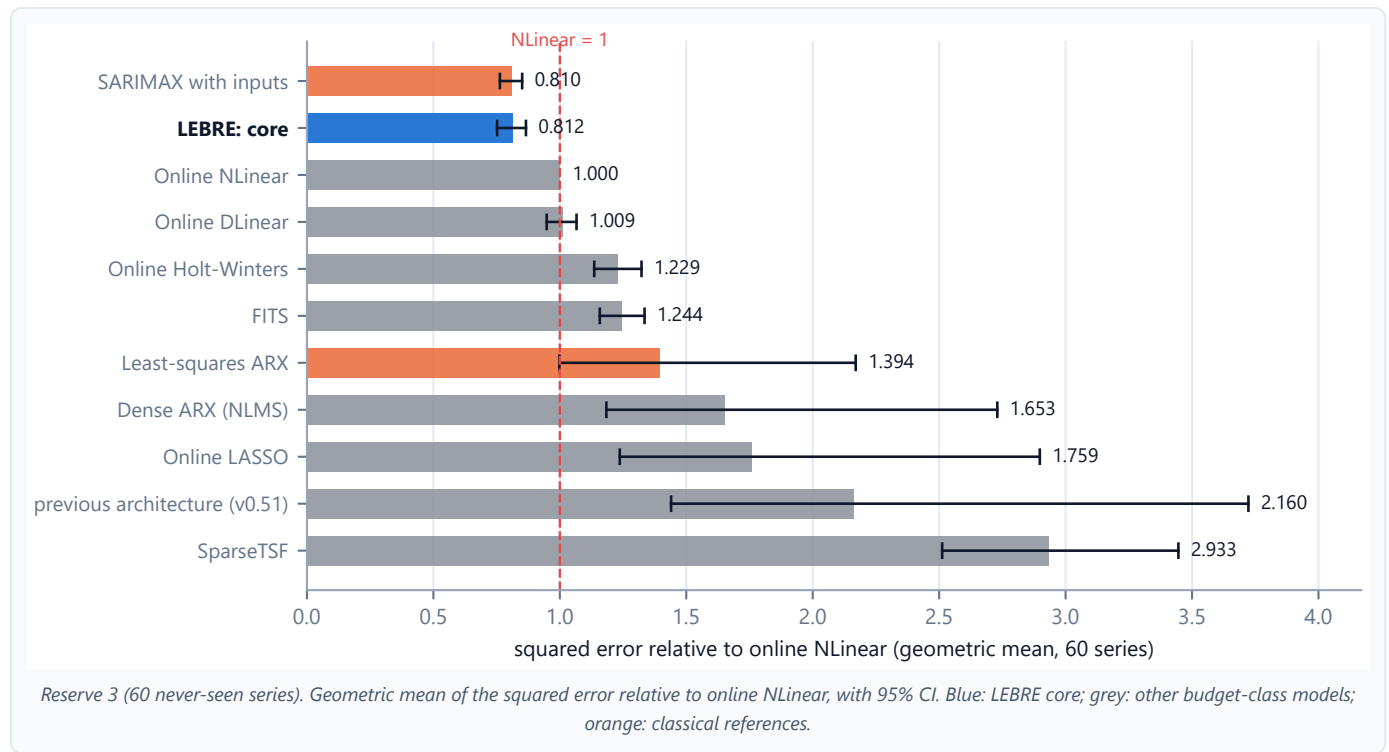


- **Nulls:** 0 of 400 runs with any accepted change.
- **With one true input:** by the declared criterion (per scenario, 100 runs) the three scenarios stayed within $\alpha = 0.05$ (FDR 0.045 and 0.038 in the worst). But, with persistence 0.5, two cells exceeded 0.05: FDR 0.090 (decoy) and 0.077 (dependent noise), with 7/50 and 10/50 runs affected.
- **Mechanism:** in all 17 affected runs, the false change was an input *correlated* with the true one, accepted before it. While the true input is not in the model, this substitute does improve the reference forecast — the test answers its own question correctly, but the answer is not structural. The periodic removal test took the substitute out in 10 of the 17 cases; in 7 of 700 runs it stayed in the final model.

What this allows us to claim: in the simulated scenarios, the procedure empirically controlled false *predictive improvements* and accepted nothing on structureless targets. It does **not** guarantee true structure: with correlated inputs, a substitute may be accepted and sometimes remain. An accepted input means “it improved the forecast of the current reference, with evidence”, not “it is the cause”.

15.3 Evaluations on never-seen series

Reserve 1 (125 series): an engineering failure. The first pre-registered evaluation, with the core still without the output safeguards, revealed that **6 of 125 series exploded** (relative error from 13 up to 10^{79}): when the target was missing for many steps, the memory filled its own history with its own forecasts. The pre-registered result was dominated by these series (geometric mean 5.20); in an exploratory view, not pre-registered, without the 6 series, the relative error was 0.706, equal to development. The failure motivated the safeguards of §11, checked in development and then on never-seen series.



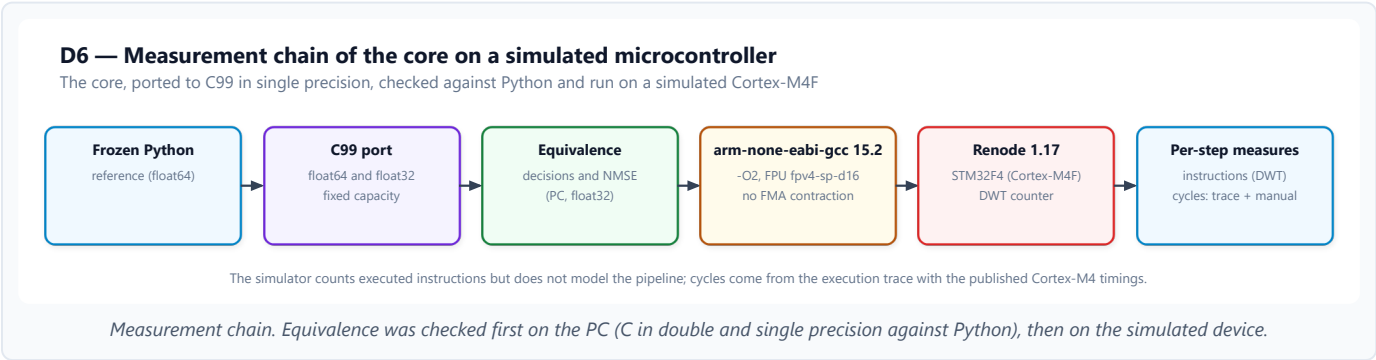
Model	CAMELS	BDG2	All [95% CI]	Series > 1.5	Catastrophic	Operations/forecast
LEBRE: core	0.858	0.768	0.812 [0.752; 0.866]	0%	0	415
Online NLinear	1.000	1.000	1.000 [1.000; 1.000]	0%	0	674
Online DLinear	1.076	0.947	1.009 [0.948; 1.066]	3%	0	1,533
Online Holt-Winters	1.213	1.244	1.229 [1.136; 1.323]	25%	0	20
FITS	1.362	1.137	1.244 [1.158; 1.335]	18%	0	9,757
Dense ARX (NLMS)	1.008	2.711	1.653 [1.184; 2.730]	38%	2	474
Online LASSO	1.015	3.048	1.759 [1.237; 2.898]	38%	4	705
previous architecture (v0.51)	2.180	2.140	2.160 [1.440; 3.723]	48%	4	108
SparseTSF	2.980	2.886	2.933 [2.512; 3.446]	87%	3	4,016
SARIMAX with inputs	0.817	0.803	0.810 [0.763; 0.851]	0%	0	ML fit
Least-squares ARX	0.677	2.867	1.394 [0.998; 2.170]	20%	4	48,944

Ceiling (1,000 points)	CAMELS	BDG2	All	Paired ratio core/model [95% CI]	Operations/forecast
SARIMAX with inputs	0.798	0.818	0.808	1.004 [0.956; 1.056]	ML fit
TTM zero-shot	1.204	0.978	1.085	0.747 [0.704; 0.795]	15.6·10 ⁶
TTM fine-tuned w/ inputs	1.049	1.054	1.052	0.771 [0.699; 0.844]	55.6·10 ⁶
Chronos-2 w/ covariates	0.782	0.539	0.649	1.250 [1.178; 1.337]	~10 ⁹ –10 ¹⁰

Summary of the core evaluations with the safeguards: error relative to NLinear of 0.767 (reserve 2, 40 series) and 0.812 (reserve 3, 60 series); no catastrophic series and none above 1.5 among the 100 series; a tie with SARIMAX with inputs in both (paired ratios 0.999 and 1.002); lower error than TTM zero-shot and fine-tuned; error 16% and 25% above Chronos-2 with covariates. The ultralight FITS and SparseTSF, batch-fitted on the first 15%, were behind online NLinear in this one-step regime, for which they were not designed. The 90% interval had empirical coverage of 0.905 and 0.904.

16. The core on a simulated microcontroller

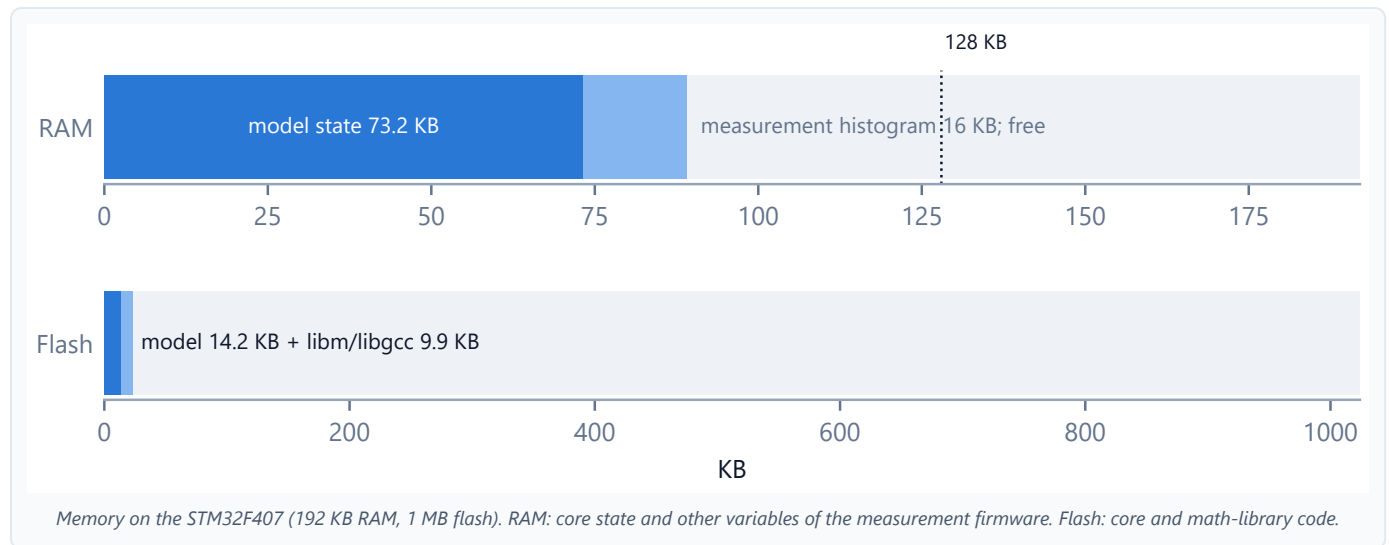
To check whether the analytic cost corresponds to something executable on a microcontroller, the frozen core was ported to C99 and run on a **simulator** of a common microcontroller (STM32F4, Cortex-M4F core at 168 MHz). There was no measurement on a physical board. **M1 and M2 were not ported**: for them there is only the analytic count (§12).



Equivalence. In double precision, the C code exactly reproduces the Python decisions on 29/29 development series. In **single precision** (the microcontroller build), the sequences of accepted changes are identical on 28/29 development series and on 54/60 never-seen series; on the others, a decision near the threshold happens a few steps earlier or later (rounding), with a median error difference of $\sim 10^{-6}$. **Cost.** The simulator (Renode 1.17) counts executed instructions via the DWT counter, but does not model the pipeline; cycles were estimated from execution traces with the published Cortex-M4 and FPU timings: CPI 1.62 and 1.57. Times below use CPI 1.6 and are estimates.



Series	Source	Inputs	Mean	Median	p99.9	Peak	Time	NMSE	Changes
ONS (dev.)	dev.	1	3,703	3,363	12,327	42,021	~35 μ s	0.2826	4
CAMELS (dev.)	dev.	3	4,560	4,599	14,387	48,447	~43 μ s	0.0229	3
BDG2 (dev.)	dev.	2	5,964	5,607	16,843	45,654	~57 μ s	0.0305	2
CAMELS 51795000	reserve 3	3	3,203	3,003	12,223	36,771	~31 μ s	0.1465	0
CAMELS 64625000	reserve 3	3	3,521	3,215	12,267	36,708	~34 μ s	0.3249	0
BDG2 Fox education Ollie	reserve 3	2	5,806	5,587	15,375	45,402	~55 μ s	0.0162	3
BDG2 Panther education Vincent	reserve 3	2	5,528	5,463	16,971	45,696	~53 μ s	0.0584	3



Reading: ~3.2–6.0 thousand instructions per step correspond to ~9–15 instructions per analytic operation (memory accesses, loops, indices, divisions). At ~31–57 μ s per step, a 1-second sampling would take < 0.01% of the processor core time; energy was not measured. With M1 and M2, which roughly double the analytic count (§12), the expected order of magnitude stays the same, but it **was not measured**. The state fits the STM32F407 RAM, but not smaller microcontrollers without reducing capacities.

17. Development of M1 and M2

A public test bench (the LEBRE Lab) ran the frozen core on new domains and surfaced failure modes. M1 and M2 address two of them, without widening the core's hypothesis space; the others were left outside this version (§22).

Failure mode 1: losing to the trivial reference (motivates M2)

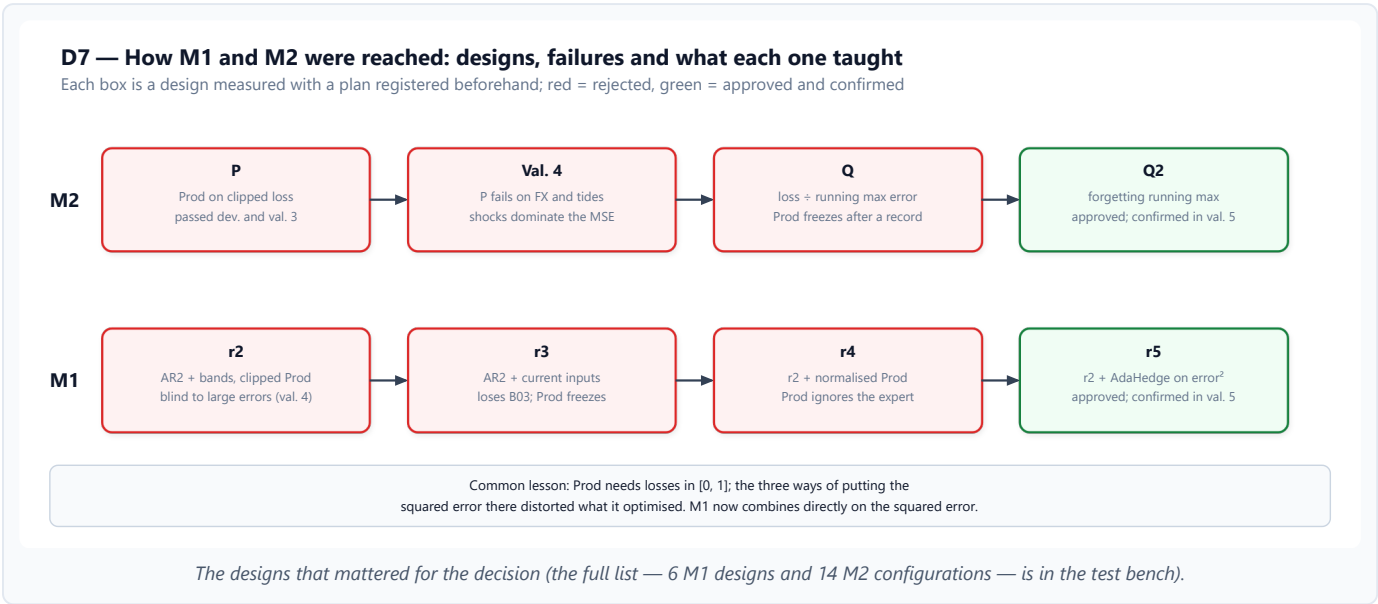
In nearly unpredictable series, the core's adaptive part errs more than the trivial reference. In the final reserve itself, the reference beat the core, with the whole confidence interval above 1, in 5 of 8 FX return series and 6 of 8 level series.

Failure mode 2: below the accuracy × cost frontier (motivates M1)

In 8 development families a full linear comparator (all inputs and lags estimated jointly, from thousands to tens of thousands of operations per step) beats the core by more than 5%. What matters under a budget is whether the core loses to something *of the same cost*. The diagnostic measured 8 regressions of varied cost in each family: the core is below the frontier (a regression of equal or lower cost errs 5% less) in 4 of them, all with the same cause, the lack of joint estimation of the target dynamics and of the input effect.

Family	core ÷ full	best with cost ≤ core	core cost (FP)	status
B03	2.076	1.574	731	below
VB07	1.170	1.005	435	below
XB07	1.141	1.010	417	below
B07	1.067	1.004	431	below
YB06	1.095	1.048	640	on the frontier
YB03	1.225	1.397	830	on the frontier
VC05	1.138	1.381	262	on the frontier
A05	1.077	1.777	385	on the frontier

Error = MSE ÷ MSE of the full online linear comparator (family geometric mean). B03: air quality; B07, VB07, XB07: storm surge (three independent sets); VC05: inflation; YB03: Beijing air quality; YB06: wind; A05: synthetic.



Family	core ÷ full	LEBRE ÷ full	target	cost ÷ core
B03	2.076	1.365	1.652	2.01
B07	1.067	1.020	1.054	2.17
VB07	1.170	1.053	1.056	2.13
XB07	1.141	1.044	1.061	2.21

Process findings, all recorded: a numerical defect in path W (0/0 by *underflow*) was found in a test, fixed with the log-scale fallback and audited (it never fired in a recorded measurement; since then a non-finite forecast counts as a failure); a construction error in validations 4 and 5 (30 h cycle instead of 24 h in solar, wind and load) restricted the scope of validation 5 (§18).

18. Validations on new data

Validations 1 and 2 served earlier M2 designs; validation 3 confirmed candidate P. **Validation 4** rejected M1 draft 2 with P: criterion 1 on YB07, YC01 and criterion 2 on YB01. The diagnostic separated the causes (the solar failure came from M1; the FX and tide ones from M2) and led to the designs of §17.

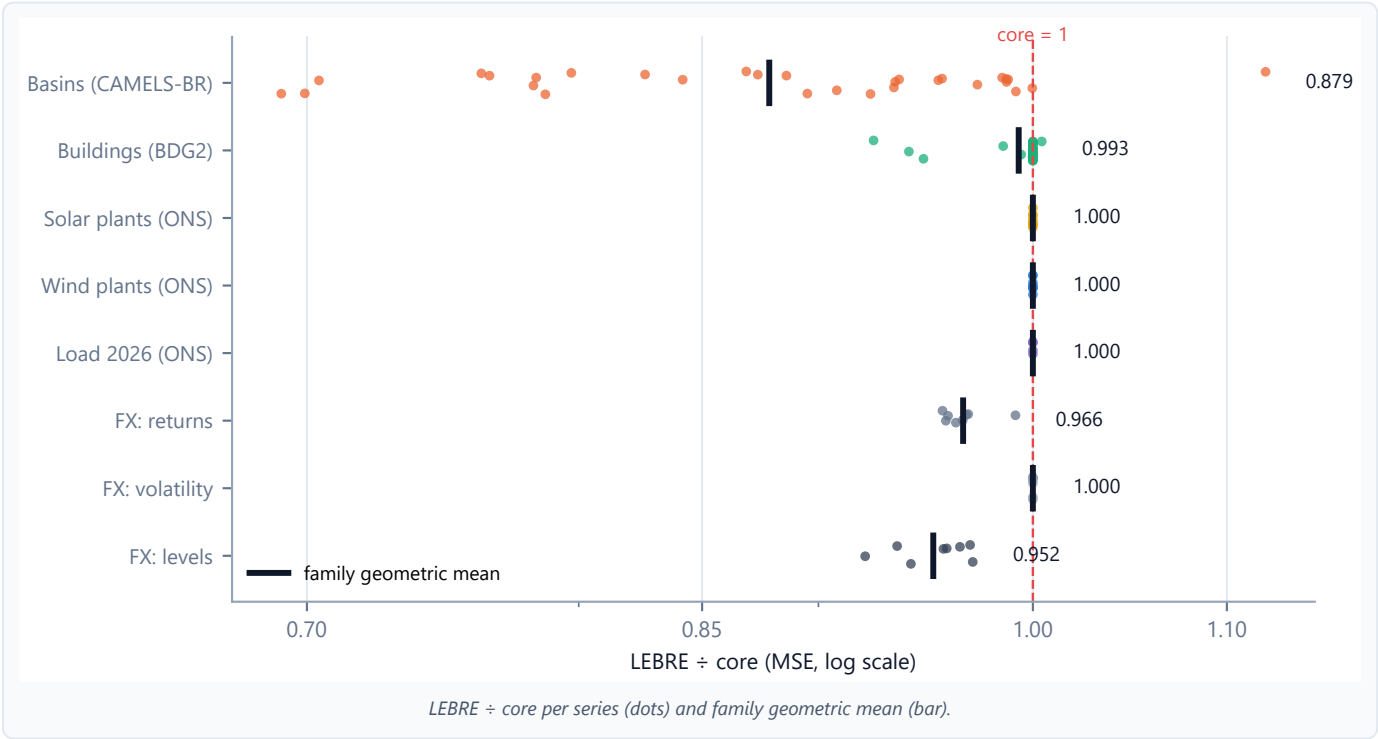
Validation 5 (43 new series, single measurement) confirmed M1 draft 5 with M2 Q2: F6 met on Beijing air quality (Chen, 2017) and on six new tide stations, and all M2 criteria on every family.

Family	core ÷ full	LEBRE ÷ full	target	cost ÷ core
Z-B03	1.169	1.118	1.410	1.97
Z-B07	1.017	0.925	1.068	2.20

Errata: in validations 4 and 5, the solar, wind and load families were built with a 30 h cycle instead of 24 h. The core vs LEBRE comparisons in those families hold for the task as built, not for the declared analogue; the fresh evidence for LEBRE with the right cycle in those domains is the final evaluation (§19).

19. Pre-registered final evaluation

Single run on 104 reserved series (30 CAMELS-BR basins; 30 BDG2 meters; 8 solar plants, 8 wind plants, the 2026 load of the 4 ONS subsystems and 8 FX pairs in 3 tasks: returns, volatility and levels). The pre-registration was approved after a rehearsal on 55 already spent basin and building series (0 errors). Result: **promoted** by the rule fixed beforehand.



Family	series	core better / ref. / undec. / shock	LEBRE ÷ core	crit. 1	crit. 2	max +cost (FP)
Basins (CAMELS-BR)	30	15/2/7/6	0.879	0.845 [0.814; 0.878]	0.911 [0.899; 0.924]	495
Buildings (BDG2)	30	30/0/0/0	0.993	-	0.993 [0.988; 1.000]	385
Solar plants (ONS)	8	8/0/0/0	1.000	-	1.000 [1.000; 1.000]	696
Wind plants (ONS)	8	8/0/0/0	1.000	-	1.000 [1.000; 1.000]	918
Load 2026 (ONS)	4	4/0/0/0	1.000	-	1.000 [1.000; 1.000]	831
FX: returns	8	0/5/1/2	0.966	1.000 [1.000; 1.000]	-	894
FX: volatility	8	6/0/0/2	1.000	-	1.000 [1.000; 1.000]	894
FX: levels	8	0/6/0/2	0.952	1.001 [1.000; 1.002]	-	894

- Overall improvement: **0.955 [0.951; 0.960]**; short series against the best of core and R: 1.007 [0.995; 1.017].
- No non-finite forecast; false inputs on returns equal to the core's; structure preserved on every series; estimated memory 85,092 B.
- In solar, wind and load, M1 ended with zero weight on all 20 series and LEBRE was identical to the core.
- Two fixes during the run, with no effect on model or criterion: reading the 2026 load by subsystem code (ONS renamed the Southeast) and handling a degenerate short series (exact tie with zero error).

Reference ceilings (reported only)

Models outside the budget class, as reference only: Chronos-2 with covariates (Ansari et al., 2025), on the same 1,000 points per series as protocol (a); and SARIMAX with inputs, fitted on the first 15%.

Family	LEBRE ÷ Chronos-2	core ÷ Chronos-2	SARIMAX-X ÷ LEBRE
Basins (CAMELS-BR)	0.885	1.008	1.134
Buildings (BDG2)	1.248	1.241	1.034
Solar plants (ONS)	1.237	1.237	1.096
Wind plants (ONS)	1.051	1.051	1.049
Load 2026 (ONS)	1.433	1.433	0.968
FX: returns	0.954	0.979	1.123*
FX: volatility	0.983	0.983	1.058*
FX: levels	0.881	0.921	1.065*

* FX: SARIMAX with inputs did not run in the execution (inputs with missing holidays); values from an addendum made after the result, with inputs filled by the last value.

20. Evolution of the architecture and ablations

Evolution

LEBRE evolved through frozen versions, each with its hashes. v0.1 (the project's canonical reference) defined the idea: structure that enters and leaves the model by evidence, within a budget. v0.51 used two experts with per-lag discovery and martingale/CUSUM tests; later analyses showed that, on real series, almost all of the result came from the memory. v0.52 kept the memory and the combination and replaced discovery by a generic engine of tested changes — it is this document's **structural core**, frozen on 2026-09-27 without promotion. v0.53 (this document) added M1 and M2 without changing a byte of the core and was promoted on 2026-10-09. The internal steps of the core were:

- per-lag hypotheses (~400) → **per whole input** (fewer hypotheses; better on real data);
- bands diluted isolated delays → **peak lag** and hierarchical splits;
- slow evidence → **one-sided c** and persistent peak statistic;
- acceptances on flat stretches → loss **scale floor**;
- acceptances on structureless targets → **step 0.05** in the reference (§8);
- cost peaks → one warm-up at a time, spread screen, half-window peak statistic;
- explosions in gaps → **hold and output contract** (§11).

Ablations

Variant	What changes	Result
Core alone (no M1 and M2)	the output is L	final evaluation: LEBRE ÷ core = 0.955 [0.951; 0.960]; M1 has zero weight in solar, wind and load
Core: all on, no tests	all inputs and the latent state active from the start	reserves 2 and 3 (100 series): ratio 0.988 [0.970; 1.006], ~5% lower cost — exploratory; the testing mechanism does not improve accuracy distinguishably
Core without hierarchical splits	the response stays at octave bands + peak	ratio 0.996 [0.988; 1.001] — no difference on real data
Per-lag hypothesis ("atomic")	one hypothesis per lag of each input (~400 hypotheses)	development: 0.757 vs 0.645 (1,000 points); slower discovery
No output safeguards	open-loop memory during gaps	reserve 2: 1.107 vs 0.767; 2 catastrophic; paired ratio 0.693
Base with step 0.1	larger reference misadjustment	10/40 acceptances on structureless targets (§8)
Previous architecture (v0.51)	two experts with per-lag discovery, no stream control	reserve 3: 2.160, 4 catastrophic
M1 and M2 with other combiners	clipped Prod, normalised Prod, other designs (§17)	rejected in development or in validation 4

Consequence for what can be claimed: the core's accuracy comes mainly from a cheap online linear regression on the current inputs and a seasonal memory, protected by output safeguards; LEBRE's gain over the core comes from M1 and M2 (§19). The mechanism of tested changes **is not justified by an accuracy gain**; it is justified, if at all, by the structure *logged with evidence* (which inputs, with which delay, since when, with which evidence). **We did not demonstrate** that this structure improves any practical decision (diagnosis, control, maintenance).

21. Position in the literature

Aspect	Related work	Classification
Structural change accepted by an e-process of predictive improvement, with stream error control	Amoukou, Mishra & Veloso (2026), in online decision trees (FWER control); Choi (2026), prespecified predictive corrections; Jha (2026), Granger causality with sequential testing; Choe & Ramdas (2024); Xu & Ramdas (2024)	not original
Per-input units, hierarchical refinement, tested removal and swap, FDR control, fixed budget	known parts (hierarchical testing; e-LOND; persistent hypotheses with FDR: de Heide 2026; doubly sequential testing: Yao, Gang & Sun 2025)	integration/extension
Aggregation of forecasters (M1, M2)	AdaHedge (de Rooij et al. 2014); (A,B)-Prod and constant regret against a trusted expert (Sani et al. 2014; Even-Dar et al. 2008; Koolen 2013); switch distribution (van Erven et al. 2012) and Fixed Share (Adamskiy et al. 2016; Cesa-Bianchi et al. 2012); recursive regression with lags (Ljung 1999)	not original; choice of losses and chaining
Unbounded and heavy-tailed losses	Cesa-Bianchi et al. (2007); Hazan & Kale (2010); Orabona & Pál (2018); Mhammedi et al. (2019); V'yugin & Trunov (2019); Alquier (2021); Moulin et al. (2025); robust confidence sequences (Wang & Ramdas 2023), future direction	known limit; handled in the ruler
Condition of use (reference misadjustment)	Giacomini & White (2006); Clark & West (2007); LMS misadjustment (Widrow & Stearns 1985); Han & Qu (2026); incumbent bias in challenger promotion (Fernández-Barrios et al. 2026)	modest methodological observation
Forecasting with a budget of $\sim 10^2\text{--}10^3$ operations per step	SparseTSF, FITS, TTM (generally univariate and batch-trained); embedded forecasting with incremental learning and exogenous sensors (Lara Pinal, Das & Schlüter 2026)	empirical contribution, within the tested scope
Hold, output contract, cost limits	engineering practice	standard

The literature audit of the core followed a protocol with claims and queries fixed beforehand (47 queries, in two rounds) and classified each work as anticipation, partial, basis or distant; no work found anticipates the whole combination, and each part has a precedent. **Limitations:** one general search engine, English queries, reading mostly by abstract, no double screening; it is not a systematic review in the formal sense. LEBRE's contribution is the **integration** of these components under a declared budget and the pre-registered evaluation.

22. Limits and claims we do not make

- **Horizon:** one step ahead only, with the current inputs available.
- **Domains:** the core was evaluated against the budget class on hydrology and buildings; M1 and M2, on basins, buildings, solar, wind, load and FX, plus air quality and tides in the validations. Other domains were not tested.
- **Cost without gain in part of the domains:** M1 and M2 cost on every series; in solar, wind and load LEBRE is identical to the core.
- **Core cost contract** not met (3.7% in the mean and 2.0% in the peak), without later revision; the M1 + M2 ceiling was redefined before the final evaluation (§12).
- **Core testing mechanism:** does not improve accuracy; false-change control is in the practical regime, not in the exact-guarantee one (§8); inputs correlated with the true one may be accepted as substitutes and sometimes remain (§15).
- **Discovery:** fails on very smooth inputs with medium/long delays; near-identical inputs are indistinguishable.
- **No formal guarantee in the original scale for M1 and M2:** Prod guarantees on normalised losses; M1's AdaHedge gives no constant regret against the core; with shocks, no combiner has a useful guarantee (§6.4).
- **Ruler decided during development:** the shock addenda were decided after seeing development data, before validation 5 and the final evaluation.
- **F6 without a family in the final reserve:** the evidence for F6 on new data is validation 5 only; one development family passed by a small margin (VB07).
- **Microcontroller:** core only, simulator only, cycles estimated, energy not measured; M1 and M2 memory estimated.
- **Uncertainty:** the overall CIs treat series as independent; load has 4 series; 4 BDG2 sites share weather with the core's development.
- **Missing comparators:** an online ARMAX with a Kalman filter and a SARIMAX with online parameter updates.
- **Interval:** empirical coverage near 90% in the core, without a formal guarantee; not reported for the complete output.
- **Post-failure fixes:** the safeguards and the final M1 and M2 designs came after seeing failures; they were confirmed on new data, but the design was not blind. Errata and run fixes are declared.
- **Outside this version:** removals that actually happen, cost proportional to use and the cycle treated as a hypothesis.

We do not claim that LEBRE is the best forecaster, nor the most economical, nor that any component is new. We only claim what was measured: under the protocols and data described, at the cost described, the numbers reported in §15–§19.

23. Parameters

All constants are the same for any series; nothing is tuned per series.

Core	Constants
Units	one per input (includes the target's own past) + residual latent state (poles 0.8 and 0.95); up to 4 active
Response	bands [1], [2–3], [4–7], [8–15], [16–31] + peak lag (chosen at 200 samples) + current value; splits into halves
Challengers	RLS within episode, $P_0 = 10^{-1}$; warm-up 250; 2 slots; one warm-up at a time
Evidence	loss $\min\{e^2/B^2, 1\}$, $B = 2 \cdot \max(\hat{\sigma}_S, 0.1 \cdot s_{\text{slow}})$; $c = 1 + \max(\epsilon, 0)$; $8 \lambda = 0.9 \cdot 2^{-1/c}$; $\epsilon = +0.002$ (add, swap, split), -0.002 (remove)
Multiplicity	$\alpha = 0.05$; $\gamma = 1/(2P)$ for the first $P = 2 \cdot \text{units} $, then $0.5 \cdot \gamma_k \propto 1/(k \log^2 k)$; level $\alpha \cdot \gamma \cdot (R+1) \leq 0.5$
Episodes	scan after 100 samples; limit 5,000; wait 2,000; periodic removal every 2,000; decisions every 10 steps
Screening	prewhitened AR(1), weight 0.02, every 4 steps per part; thresholds 15.09 (χ^2_s) and 6.63 (χ^2_1)
Base of S	NLMS $\mu = 0.05$; input contract $\pm 8\sigma$ with a 31-step quarantine
Memory	NLMS $\mu = 0.05$; slow mean 0.01; profile 0.1; floors 10^{-3} and 10^{-4} ; declared cycle(s); hold during gaps
Combination and output	$\lambda = 0.99$, $\eta = 0.5$, every 8 steps; envelope with rate 10^{-3} and margin $0.5 \times$
Interval	$\alpha = 0.1$, $\gamma = 0.1$, every 8 steps (also on the final output)
Groups	causal correlation ≥ 0.95 reported with the accepted input (does not affect decisions)

Part	Parameter	Value	Note
M1	inputs	$m = \min(d, 5)$	largest absolute correlation (exponential means 0.01, past only)
M1	bands	[0], [1], [2–3], [4–7], [8–15]	means; + target at lags 1 and 2; $k = 3 + 5m$
M1	RLS	$\lambda = 0.999$; $\delta = 100$	every 8 learned targets; E only after k updates
M1	selection	after 100; every 5,000	restarts regression and combination
M1	clip	$L \pm 2 \cdot \max(\hat{\sigma}_L, \text{floor})$	floor: $0.1 \times \text{slow target scale}$
M1	combination	AdaHedge	squared error; 100-update shadow
M2	R	zero / seasonal / persistence	declared with the data
M2	H	AdaHedge	squared error, only with R defined
M2	W	one-way switch, hazard $1/(n+1)$	Gaussian losses at the recent scale (0.01)
M2	Prod	$\eta \leq 1/2$; $w_B = 1/2$	losses $\div \max(\text{losses}, 0.99 \cdot N)$
M2	gate	$\epsilon = 0.002$; $\alpha = 0.01$	audit only; recreated after each decision

24. Pseudocode and reproducibility conditions

Simplified pseudocode of one step, sufficient to re-implement the evaluated configuration. Where text and pseudocode differ, the pseudocode together with the parameter table (§23) prevails.

Structural core

```

CONSTANTS # the same for any series; nothing is tuned per series
BANDS = [1], [2-3], [4-7], [8-15], [16-31]      POLES = 0.8, 0.95      L = 31
M_MAX = 4    SLOTS = 2    WARM = 250    PEAK_N = 200    N_MIN = 100    T_MAX = 5000    COOL = 2000    REM_PERIOD = 2000
DECIDE_EVERY = 10    SCREEN_EVERY = 4    SCREEN_W = 0.02    Q_UNIT = 15.09 ( $\chi^2_s$ , 99%)    Q_SPLIT = 6.63 ( $\chi^2_1$ , 99%)
 $\epsilon$ : add = swap = split = +0.002, remove = -0.002     $\alpha$  = 0.05     $\mu_S$  = 0.05     $\mu_M$  = 0.05    CLIP_X = 8
 $\lambda_{DMA}$  = 0.99     $\eta$  = 0.5    EVERY = 8     $\alpha_{int}$  = 0.10     $\gamma_{int}$  = 0.10    SCALE_FLOOR = 0.1    OUT_MARGIN = 0.5    OUT_RHO = 1e-3

STATE # fixed size, given the number of inputs d and the period(s) s, s2
H[0..L+1][d+1] input history (includes the target's own past, standardised)
band_sum[5][d+1], seg_sum{(i,lo,hi)} running sums of the bands and of active segments
base[d+2] active{unit → (segments, weights)} r[2] residual latent states
M: buffer, w_M, G[s], m_slow      hyps{key → e-process}      slots[≤ 2]

STEP(x_t, y_t or gap, quarantine):
# 1. read
 $\bar{x} \leftarrow [x_t, \text{std\_causal}(y_{\text{last}})]$  # EMA mean/variance 1e-4, start (0, 1)
if any  $|\bar{x}| > \text{CLIP\_X}$ : quar_until ← t + L;  $\bar{x} \leftarrow \text{clip}(\bar{x}, \pm \text{CLIP\_X})$ ; shift H, band_sum, seg_sum
# 2. predict
 $\hat{y}_M \leftarrow M.\text{predict}()$  # last value +  $w_M \cdot \phi$  (reversion, difference, seasonal, profile)
 $\hat{y}_S \leftarrow \text{base} \cdot [1, \bar{x}] + \sum_u w_u \cdot \text{block}(u)$  # block: means of the active segments
 $\hat{y} \leftarrow \hat{y}_M + w_S (\hat{y}_S - \hat{y}_M)$ 
if n_obs ≥ 50:  $\hat{y} \leftarrow \text{clip}(\hat{y}, \text{env\_lo} - 0.5 \cdot W, \text{env\_hi} + 0.5 \cdot W)$ ,  $W = \text{env\_hi} - \text{env\_lo}$ ; if  $\hat{y}$  not finite:  $\hat{y} \leftarrow y_{\text{last}}$ 
for each experiment e:  $g_e \leftarrow \hat{y}_S + \theta_e \cdot z_e - (\text{removed contribution})$ 
return  $\hat{y}$ , [ $\hat{y} - \hat{q}$ ,  $\hat{y} + \hat{q}$ ]
# 3. observe and learn (only if y_t is observed)
learn ← y_t observed  $\wedge$  ¬quarantine  $\wedge$  t > quar_until
 $B \leftarrow 2 \cdot \max(\sigma_r, \text{SCALE\_FLOOR} \cdot \text{sd\_slow}(y))$ ;  $\ell(u) = \min((y_t - u)^2 / B^2, 1)$ 
if learn:
  for each e: k_e += 1
    if k_e > WARM: e.hyp.observe( $\ell(\hat{y}_S) - \ell(g_e) - \epsilon_e$ ) # evidence
    if k_e ≤ PEAK_N: accumulate prewhitened correlation per lag (half the lags per step)
    if k_e = PEAK_N:  $k^* \leftarrow \text{argmax}_k c_{xy^2}/c_{xx}$ ; if  $k^* > 1$ : insert  $x_i(t-k^*)$  into the challenger
    RLS( $\theta_e, P_e$ ;  $z_e, \text{alvo} = y_t - \hat{y}_S + \text{removed contribution}$ ) #  $P_0 = 10 \cdot I$ 
    NLMS(base, w_u; res =  $y_t - \hat{y}_S, \mu_S$ , norm =  $\| [1, \bar{x}] \|^2 + \sum \|\text{block}(u)\|^2$ )
    screen_spread(res) # each part on its own phase, every 4 steps
   $D \leftarrow \lambda_{DMA} \cdot D + ((y_t - \hat{y}_S)^2 - (y_t - \hat{y}_M)^2) / \sigma^2$ ; every EVERY:  $w_S \leftarrow 1 / (1 + \exp(\eta \cdot D))$ 
  interval: miss <  $|y_t - \hat{y}| > \hat{q}$ ; every EVERY:  $\hat{q} \leftarrow \max(0, \hat{q} + \gamma_{int} \cdot \sigma \cdot \Sigma(\text{miss} - \alpha_{int}))$ 
  M.update(y_t if observed; otherwise the last observed value (hold), without learning)
  env ← envelope of observed values (max/min relaxing to the mean, rate OUT_RHO)
  r ← POLES · r + (1 - POLES) · (y_t -  $\hat{y}$ ) /  $\sigma$ 
  if t mod DECIDE_EVERY = 0: DECIDE()

DECIDE():
  for each e: if log E(e.hyp) ≥ log(1/ $\alpha_{hyp}$ ): ACCEPT(e)
    else if n_e ≥ T_MAX or (n_e ≥ N_MIN and  $\sum \text{episode } d \leq 0$ ): close (the evidence stays with the hypothesis)
  close experiments about units that just changed; FILL()

FILL(): while a slot is free:
  1. active unit without a removal test for ≥ REM_PERIOD → open remove
  2. if some challenger is warming up: stop # one warm-up at a time
  3.  $u^* \leftarrow \text{argmax stat}(u)$  among inactive, free units out of cool-down;  $s^* \leftarrow \text{argmax stat}(splits)$ 
  4. pick the larger of stat( $u^*$ )/Q_UNIT and stat( $s^*$ )/Q_SPLIT (if ≥ 1):
    split → split; |active| < M_MAX → add; else → swap (with the unit of smallest contribution)
  # a new hypothesis n receives a fixed level:  $\alpha_n = \min(\alpha \cdot \gamma_n \cdot (R+1), 0.5)$ ,
  #  $\gamma_n = 1/(2P)$  for  $n \leq P = 2 \cdot |\text{units}|$ , then  $0.5 \cdot C / ((n-P) \cdot \log^2(n-P+1))$ ; R = changes accepted so far

e-process( $\epsilon$ ): c = 1 + max( $\epsilon$ , 0);  $\lambda_j = 0.9 \cdot 2^{(-j)}/c$ , j = 0..7;  $\psi_j = (-\log(1 - c\lambda_j) - c\lambda_j)/c^2$ 
  observe(d):  $y \leftarrow \text{clip}(\text{past mean of } d, -c, 0)$ ; S += d; V += (d - y)2
  log E = log( (1/8) ·  $\sum_j \exp(\lambda_j \cdot S - \psi_j \cdot V)$  )

ACCEPT(e): add/swap → active[i] ← (BANDS  $\cup \{[k^*]\}$ ,  $\theta_e[-1]$ ); base[i] +=  $\theta_e[-1]$ ; remove → delete active[u]
  split → replace band [lo,hi] by its two halves with the challenger weights; R += 1

```

M1 and M2 (on top of the core forecast L)

```

M1 and M2 CONSTANTS  M_MAX = 5  BANDS_E = [0], [1], [2-3], [4-7], [8-15]  λ_RLS = 0.999  δ_RLS = 100
                      EVERY = 8  CLIP_K = 2  N_MIN = 100  T_MAX = 5000  LAM = 0.99  SCALE_FLOOR = 0.1  ε_gate = 0.002  α_gate = 0.01

PREDICT(x_t):
  L ← core.predict(x_t)                                # core pseudocode above
  # M1: precision expert
  x ← x_t with missing values replaced by the last value seen
  z ← [1,  $\tilde{y}_t(t-1)$ ,  $\tilde{y}_t(t-2)$ , BANDS_E means of the m = min(d, M_MAX) chosen inputs, standardised]
  E ← clip(w·z, L ± CLIP_K·max(σ_L, floor))             if RLS determined ( $n_{\text{RLS}} \geq k = 3 + 5m$ )
  L' ← L if E undefined or AdaHedge1.n < N_MIN; else (1 - w)·L + w·E      # shadow
  # M2: start from the trivial reference
  R ← 0 | y_last | y_t(t-s)                            # zero, persistence or seasonal naive, declared
  if R undefined:  $\hat{y} \leftarrow L'$ 
  else: H ← w_H·[R, L']; W ← w_W·[R, L']; s ←  $\eta \cdot w_A / (\eta \cdot w_A + w_B/2)$ ;  $\hat{y} \leftarrow s \cdot W + (1 - s) \cdot H$ 
  return  $\hat{y}$ , [ $\hat{y} - \hat{q}_o$ ,  $\hat{y} + \hat{q}_o$ ]

OBSERVE(y_t, quarantine):
  core.observe(y_t, quarantine)
  if y_t observed:
    AdaHedge1.update((y_t - L)2, (y_t - E)2)          # squared error, no clipping
    learn ← ¬quarantine ∧ inputs within the contract
    if learn: every EVERY targets: screening (exponential means 1 - LAM) and RLS(z, y_t)
               selection of the m inputs after N_MIN targets and every T_MAX; if it changes: RLS and AdaHedge1 restart
    if ¬quarantine ∧ R defined:
      AdaHedgeH.update((y_t - R)2, (y_t - L')2)
      s2 ← max(min(σR2, σL'2), floor2); Switch_W.update((y_t - R)2/2s2, (y_t - L')2/2s2)  # log-scale fallback against 0/0
      N ← max(LAM·N, (y_t - H)2, (y_t - W)2); Prod.update((y_t - H)2/N, (y_t - W)2/N)
      gate.observe(...)                                # audit: logs events, does not decide the output
    output interval: miss ← |y_t -  $\hat{y}$ | >  $\hat{q}_o$ ; every EVERY:  $\hat{q}_o \leftarrow \max(0, \hat{q}_o + 0.1 \cdot \sigma_o \cdot \mathbb{I}(\text{miss} - 0.1))$ 

```

Reproducibility conditions

- **Inputs:** standardise each input causally before the core: $z = (x - m)/\sqrt{v + 10^{-6}}$, with exponential m and v at rate 10^{-4} from (0, 1), updated *after* standardising the current value; missing values become 0 after standardisation (in M1, the last value seen).
- **Target:** passed in original units; missing values are passed as a gap (no learning); an external quarantine flag suspends learning.
- **Metadata:** seasonal period s (24 for hourly data; none if there is no cycle), optionally a second cycle s₂ (168 for buildings), and the trivial reference (zero, persistence or seasonal; if not declared: seasonal if there is a cycle, otherwise persistence).
- **Determinism:** the algorithm uses no random numbers; given the same input sequence, the output is deterministic.
- **Availability:** the promoted code (Python), the C port of the core, the pre-registrations, the analysis scripts, the per-series forecasts and the SHA-256 manifests of every frozen version are in the public repository github.com/Iqitim/lebre-research (code of the complete LEBRE at commit 4a2620e); the development and validations of M1 and M2 are in the public test bench github.com/Iqitim/lebre-lab (this document read its results at commit 6f328bb).
- **Environment used:** Python 3.11, numpy 2.2.5, pandas 2.2.3 (later versions changed read-only array behaviour), statsmodels 0.15.0; Chronos-2 with the chronos 2.3.2 package; C99 with Arm GCC 15.2 (−O2, single-precision FPU, no FMA contraction); Renode 1.17.

25. References

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26. Revision history

- **Revision 0** (2026-10-09): first edition, after the promotion; it described only the new components.
- **Revision 1** (2026-10-09): self-contained document — it describes the whole LEBRE (core, M1 and M2: architecture, theory, lifecycle, condition of use, safeguards, cost, evidence, parameters and pseudocode) without assuming the reading of any other document. No result was changed.